HYBRID HOME SECURITY SYSTEM WITH CONTACT DETECTION AND COMPUTER VISION–BASED FACIAL RECOGNITION

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CERTIFICATE OF ORIGINALITY

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material to which to a substantial extent has been accepted for award of any other degree or diploma of a university or other institute of higher learning, except where due acknowledgement is made in the text.

We also declare that the intellectual content of this thesis is the product of our work, even though we may have received assistance from others on style, presentation and language expression.

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DEDICATION

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BLM

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MBR

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TTT

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TABLE OF CONTENTS

Page

LIST OF TABLES

LIST OF FIGURES

LIST OF APPENDICES

Appendix **Page**

Abstract

Title: Hybrid Home Security System with Contact Detection and Computer Vision-Based Facial Recognition

Authors: Bea L. Mindanao, Moises B. Robles and Trina T. Tagulinao

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This research introduces the concept of the Hybrid Home Security System that employs both contact and facial recognition using computer vision technology to enable effective intrusion prevention. Unlike conventional security systems, which passively monitor events, the proposed system actively detects intrusion threats and responds to knocks, wiggles, and forced attempts on doors and windows. In case of any contact, the system's built-in camera turns on and determines whether the person is authorized to enter the home. The performance of the system in terms of response time, accuracy, and reliability was tested and the results indicate that it can detect the presence of a stranger within less than three seconds. The hybrid system functioned effectively right from its installation but requires restarts now and then to avoid problems such as lagging.

Keywords: home security, contact detection, facial recognition, raspberry pi, intrusion detection.

Chapter I

INTRODUCTION

A home is a place where people should always feel safe and secure. Recent studies showed that home is not just a physical structure but a place that creates a person's identity and well-being. Maricchiolo et al. (2021) explained that a sense of belonging, attachment to place, and connectedness made home more meaningful and promoted all-around of being well.

Despite this, a lot of homes were still unsafe to intruders and theft even with locks and active surveillance. The reality of this situation is that these incidents usually happen while the residents are inside their homes without even knowing it. In the Philippine context, a national survey tells that 54% of Filipinos fear that someone might break in, showing that people worry about household security (Inquirer, 2024). According to a Social Weather Stations (SWS) survey in the third quarter of 2024, 6.1% of Filipino families reported being victims of common crimes such as break-ins or robbery in the past six months.

While CCTV cameras have become common security devices in modern homes, they are useful in recording the surroundings and recognizing intruders after the incident. However, these systems help after something bad happens. They record what happens but rarely alert owners while an intrusion is taking place. The delay of notice leaves people not ready to respond to threats in real-time (Akter, Hossain, & Rahman, 2023). Therefore, there is a strong need for technological innovations that can improve home security systems.

Background of the study

Home security remains a critical concern, especially the issue of silent intrusions that happen without the resident's awareness. A main problem in many households is that unauthorized entry can happen even while the homeowner is inside the house, yet the intrusion usually remains completely unknown until the intruder hurts someone or has already left. Even though many households depend on surveillance tools like CCTV cameras, they mostly just work as recording devices. It captures footage of events but lacks the capability to distinguish between a resident and an intruder, or to provide an immediate warning.

Even if the camera is working, a member of the researcher's family still experienced an intrusion. The intruder entered unnoticed while the resident was present. Because there was no alarm, the resident walked in on the intruder and was killed. No one found out until a family member came home later. A similar case was reported in Sampaloc, Manila, where intruders broke into the house at night while residents were asleep; the CCTV only captures the incident rather than alarming the residents(Santiago, 2025). These cases demonstrate that visual recording alone is not enough for a home security system.

Analyzing these limitations, this study aims to develop a hybrid home security system that automates threat detection. The study focuses on creating a system that has appropriate sensors to detect interactions in the doors or windows and computer vision to distinguish between the identification of the residents and intruders. It also aims to assess if the system is effective and efficient in break-in situations.

Objective of the Study

This study aims to design, develop, and evaluate a Hybrid Home Security System capable of detecting unauthorized entry, identifying registered residents, and providing alerts upon the detection of unauthorized contact or an unrecognized individual.

1. To design and develop a contact detection module capable of identifying door or window interactions using an appropriate sensor.
2. To implement a facial recognition system using computer vision algorithms.
3. To develop a sound alert system that activates when unauthorized contact or an unrecognized face is detected.
4. To test and evaluate the effectiveness and efficiency of the Hybrid Home Security System.

Significance of the study

The researchers aim to create a hybrid home security alert system that can help raise awareness among families about intruders. This study is important because it uses technology that can help households respond more quickly to danger and reduce the risk of intrusions. This system will be significant to the following:

The Homeowners. This research will contribute to improving the home security system through the application of contact sensors along with facial recognition sensors. If there is any attempt at breaking into their home, they will be informed about it. Similarly, if there is any presence of strangers in their homes, they will be warned about it.

The Community. Moreover, the research shall help to foster in people an increased awareness about the possibility of any intrusion, making family members feel safer. This increased level of awareness is instrumental in allowing neighbors to watch over one another.

The Future Researchers. This research can act as a guide for other scholars who wish to conduct further studies in the area of home security monitoring system integration. This research provides insight into how various sensors, cameras, and alarms can be used in home security system design.

Scope and Limitation

This study seeks to resolve the issue of immediacy in home security systems by developing a two-stage alarm system. While at the point of entry, such as doors and windows, the sound alarm is set off only when there is a physical contact, such as knocking, shaking, pressing, or opening, that has been detected by both the sensors and the cameras and determined that the person is not among those who have been registered. If it turns out that the person in question is one of the registered residents, then no alarm will sound even if the physical contact has been detected. In case of internal observation, the security alarm is triggered when the camera inside the house detects an unknown face. The test and evaluation stages of this security system will be applied only to homes that are located on the first floor of a building. All advanced features of this project, such as cloud storage and remote monitoring over the wider internet are excluded from this study.

Definition of Terms

In order to give clarity and consistency, the following words and phrases are explained as used in this paper. These explanations are in the context of the system that will be created and analyzed.

Algorithm refers to the specific computational logic used by the facial recognition system to process video data, extract facial features, and compare them against a database to verify identities.

Awareness refers to the homeowner's ability to know right away when a break-in is happening inside their house.

CCTV refers to security devices that only record what happens inside the house for security.

Computer Vision is the technology used to process algorithms that detect faces and identify between authorized residents and intruders.

Contact Detection Module serves as the initial trigger mechanism that detects when an entry point is opened, automatically activating the computer vision system to verify the identity of the person entering.

Facial Recognition refers to the identity recognition of a person inside the house. Verifying if that person is a resident or an intruder in that house.

Home Security System refers to the system composed of a contact detection module, implementation of facial recognition, and sound alert, that work together to keep the home protected.

Hybrid denotes the combination of a physical contact detection module (sensors) and a computer vision-based facial recognition system, working simultaneously to detect intrusion and verify the identity of individuals.

Local Network is the private Wi-Fi connection inside the house that links the contact detection module, the implementation of facial recognition, and the sound alert. It allows the system to communicate and send information quickly.

Real-time means that actions and responses happen instantly, with little to no waiting time.

Chapter II

REVIEW OF LITERATURE AND STUDIES

This chapter presents related literature and studies that serve as the technical foundation for the development of the Hybrid Home Security System. It reviews local and international works concerning contact sensing technologies, computer vision for facial recognition, and automated alert systems, particularly in the integration of physical and visual detection for improved home security.

RELATED LITERATURE

Home Security Systems

Residential security has changed from manual locks to advanced intelligent automation. As stated in contemporary research (Mun & Kim, 2024), the current security infrastructure is characterized by the implementation of IoT and real-time data analysis. This shift makes technology "proactive," protecting privacy instead of just simply recording events. The second trend seen in 2026 relates to localized data processing or "Edge AI." This makes sure that all data related to security is kept private and analyzed immediately.

Sensor Fusion for Enhanced Physical Intrusion Detection

However, the main disadvantage of residential security systems is their lack of ability to detect actual threats resulting in a high false-negative rate. This means that when an intrusion happens, the system fails to detect it. For many years, most home security systems were built on just one type of detector, such as a magnetic reed switch. According to Technavio (2025), such sensors are widely popular because of their simplicity and energy efficiency. They work by closing the circuit using a magnet and triggering the alarm only when the door is opened, and the circuit is broken. Even though they are great at detecting the intrusion, they still do not give any alerts about a person trying to break in.

One solution to this problem can be the application of a technique called Sensor Fusion, according to IEEE (2025). The process of sensor fusion involves combining information from different sensors in order to obtain better comprehension. For example, in the Hybrid Home Security system, the authors combine the function of the magnetic switch with that of an MPU6050 vibration sensor.

According to Putri et al. (2025), vibration sensors are functional to conduct an early warning. The sensors detect any vibrations or physical irregularities within a matter of seconds, making sure that any activity can be prevented from happening. While older models only reacted after the door was unlocked, the hybrid solution proposed utilizes the MPU6050 sensor to ensure that the system detects vibrations in real time. By using these sensors, a system is able to determine an appropriate threshold value based on how much shaking needs to occur for the alarm to sound, making sure that the system distinguishes normal shakes from burglary attempts. As mentioned in the Fact.MR (2026) report, "layered defense" has become the most effective solution because it reduces the amount of time between the occurrence of burglary attempts and awareness of the situation.

Wireless Transmission with ESP32 Super Mini

When considering modern security systems, the use of small-sized and compact controllers becomes crucial in terms of ensuring the secrecy of the system. The ESP32 Super Mini is an extremely miniaturized controller based on the well-known ESP32 series. As stated by Circuit Schools (2021), the ESP32 Super Mini is suitable for "IoT" purposes in which size is critical. The ESP32 Super Mini has in-built Wi-Fi and Bluetooth capabilities; thus, it can transmit information from its sensors directly to the Raspberry Pi 5 without the need for any cables.

Based on research provided by Espressif Systems (2026), the board is very efficient at processing "Low-Power Sensing." For example, the ESP32 Super Mini can be mounted on a window or door to detect information from the Contact Sensor. Due to its miniature size, the controller can easily fit into a 3D-printed miniature enclosure. As such, it will be able to remain "hidden" until it is necessary to wake up the primary video monitoring system, which is the Raspberry Pi 5. This strategy supports the creation of a "Distributed Security System."

Facial Recognition and Edge Computing

Traditional security cameras are mostly passive, and the information recorded on the tapes is viewed later. However, the National Academies of Sciences (2024) notes that the development is towards "active identification", which is capable of verifying the identity of a person in order to decide whether or not he/she is a potential risk.

A hybrid home security system achieves active identity through the use of Edge Computing, as defined by Think Robotics (2025). According to Bhardwaj et al. (2025), this means that instead of transmitting data to the slow internet server, Edge Computing is achieved on Raspberry Pi 5. This not only increases the speed but also ensures privacy. Moreover, according to Think Robotics (2025), Open Source Computer Vision (OpenCV) is capable of developing low-cost cameras that have facial recognition abilities under varied lighting situations.

As pointed out by Palkar (2025), an AI-enabled Raspberry Pi is capable of recording 15-20 frames per second, which is very important for achieving high security. By comparing the facial features of the individual to a database of registered individuals' faces, this rate enables the system to detect an intruder quickly while reducing false detections caused by traditional motion-based systems.

Raspberry Pi 5

An efficient home security system should have a quick "processor" in order to perform its tasks without any delay. The Raspberry Pi 5 is faster compared to the previous models since it offers twice as fast performance. It helps to use "edge computing", which allows the Hybrid Home Security System to perform face recognition and sensor checks right on the spot without any help from the internet. It increases privacy and improves the reaction time to intrusion attempts (XDA, 2026).

Energy Efficiency and the Raspberry Pi Zero Series

Whereas the Pi 5 takes care of the “heavy lifting” for AI, other boards, such as the Raspberry Pi Zero, are designed for unique uses. The Pi Zero, according to The MagPi (2025), is preferred in situations that involve always-on monitoring but need to have minimal power consumption. From a security perspective, the utilization of the Pi Zero means that there will be a distributed system whereby several small sensors will connect to a single hub in the form of a Raspberry Pi (Raspberry Pi Ltd, 2025). This explains why the Hybrid Home Security System should utilize the Pi Zero for secondary door nodes while using the Pi 5 as its “brain.”

Raspberry Pi Camera Module 3

The security camera should have the capability to capture high-quality images in both sunny conditions and dimly lit hallways. As per the Raspberry Pi Documentation

(2025), the Camera Module 3 uses a Sony IMX708 12 MP image sensor that incorporates two key enhancements: PDAF and HDR. Autofocus allows the camera to quickly focus on an intruder's face even if they are moving, ensuring the AI gets a sharp image for verification (Pimoroni, 2025). HDR Mode allows the camera to balance shadows and bright lights.

Security and Power Saving

The current trend for security systems is to cease continuous recording in order to conserve energy and storage capacity. As mentioned in IJSAT (2025), employing a Passive Infrared (PIR) sensor such as the HC-SR501 means that the security system can stay in a dormant state until there is detection of human activity. The PIR sensor detects changes in the amount of infrared radiation (heat) around it.

In the Hybrid Home Security System, the HC-SR501 acts as the primary "wake-up" trigger. As explained by DroneBot Workshop (2024), this creates a "half-asleep" state for the Raspberry Pi 5 and the Camera Module 3. The camera and facial recognition software only activate either upon detection of movement through the PIR sensor or by activation of the contact sensor. This system design adheres to the "Green Security" trend within the security industry that gained traction in the year 2026, which emphasizes energy-saving while ensuring adequate safety measures (SIA, 2024).

Battery Management and Power Regulation

Portable security systems need to have an energy storage device that provides a reliable and efficient power source. As explained by Bhowmick (2022), the ideal solution for high-energy storage on a small scale is the Lithium-ion Battery; however, it must be properly protected. For this reason, a BMS 1S 12A Battery Management System is incorporated into the circuit to keep track of the performance of the battery and prevent overcharge, overdischarge, and shorts in the battery. The TP4056 Module acts as a charger module that charges the battery in a safe manner following Constant Current/Constant Voltage charging. The TP4056 allows recharging through a USB port.

Voltage Regulation and Power Boosting

The major problem of using battery-powered electronics lies in the fact that the voltage generated by the battery (3.7V) is lower than that required by the Raspberry Pi 5 (5V). Texas Instruments (2026) states that Step-up/boost converters must be used to address this challenge. Modules like the MT3608 and the highly efficient TPS61088 converter are designed to raise the battery voltage to 5V. This is necessary because the Raspberry Pi 5 would fail to work properly should the voltage be reduced. As stated by Moshiri (2025) in Boost Converter News & Articles, the use of high efficiency converters such as the TPS61088 means minimal energy loss in the form of heat. The result of this is extended operational time for the Hybrid Home Security System at an acceptable speed of facial recognition operation.

Battery Monitoring and User Interface

For security purposes, there must be certainty regarding the amount of charge left in the battery. As explained by SparkFun (2025), the MAX17043 microcontroller acts as a "Fuel Gauge" for the battery. While a voltage meter provides only an estimated percentage of the battery left, the MAX17043 works by applying an algorithm to give an accurate calculation of the percentage of the battery. The information is sent to the Raspberry Pi, which in turn displays it on the 7-inch Touchscreen. Based on the documentation of the Raspberry Pi (2025), the touch screen used in this device is DSI, and therefore offers high-speed communication with lower power. This touchscreen becomes the "Control Center" of the Hybrid Home Security System, a place where users can view the remaining battery capacity, camera feed, and manage their security options without using a separate computer.

Lighting for Low Light Identification

For the facial recognition to be successful, a system needs to have good vision capabilities even in low light. As per IJAEM (2026), one of the major hurdles in AI safety systems is lighting variation, where poor images cannot be used by AI technology due to inaccurate measurement of the face landmarks. To achieve this, the Hybrid Home Security System makes use of the 1W LED Bead to provide sufficient lighting. Different from normal small LEDs, the use of a 1-watt bead guarantees adequate illumination of a face from a distance of several feet.

This makes sure that sufficient facial information is captured by the camera, which will then enable the AI algorithm to make an accurate comparison (Ingenia, 2024). In order to regulate the high-powered LED bead, the LD06AJSA LED Driver is applied. SYS Technology (2025) notes that any high-powered LED beads risk overheating or even burning up when fed with excessive currents.

Audio Alert Systems and Power Amplification

The security system will generate an audible sound that is sufficiently loud to wake up the owner of the house and frighten the burglar. The Raspberry Pi does not have sufficient power to drive an 8-ohm 3W speaker. For this reason, an amplifier is needed. For our project, we will use the PAM8403. Diodes Incorporated (2025) indicates that the PAM8403 is a Class-D audio amplifier with 3 watts. Unlike older Class-AB amplifiers that waste a lot of energy as heat, the PAM8403 is up to 90% efficient, which is critical for systems that need to stay on 24/7.

One characteristic of the amplifier that should be noted is the filterless architecture. According to ReversePCB (2025), this design enables the speaker to be driven without the need for extra bulky components, thus freeing up valuable space within the system enclosure. The use of such an amplifier guarantees that the audio signal is of high quality, enabling the "immediate awareness" required for an efficient security response.

ARCFACE

As discussed by Deng et al. (2022), ArcFace utilizes a new loss function known as the “Additive Angular Margin Loss” in order to improve the discriminating ability of the algorithm. In essence, this method creates a hypersphere wherein the “Authorized” faces tend to be grouped closer together while the “Intruders” are placed farther away. This will ensure that look-alike faces can easily be distinguished from one another.

SCRFD

As per Guo et al. (2021) in their paper released via InsightFace, SCRFD is a revolutionary algorithm in terms of efficient face detection. The study centered around "Computation Redistribution," enabling the algorithm to efficiently perform face detection at different scales (near or far) without hampering the computation process. As per the study results, SCRFD significantly outperforms traditional mobile-level detectors in both speed and mAP (mean Average Precision).

Placement for User Privacy and Immediate Response

It is important to consider the location of the security hub. According to ACM Digital Library (2025), one of the emerging trends in "Privacy-by-Design" is positioning the main control panel in a secluded place, such as the bedroom, instead of locating it near the entrance. This will ensure that in case an intruder manages to gain access to the house, he or she will have difficulty in reaching the main hub (Raspberry Pi 5) before it completes its task of alerting other users (ACM, 2025).

In addition, when it comes to emergency alarms, the user should be the first person to receive notifications, but from a safe space. In this case, the 7-inch Touchscreen hub can be located in the bedroom so that when an alarm goes off, the user can view the camera feed from a secure place without needing to leave his or her room to check if anyone has entered the house. This is consistent with the "Safe Zone Architecture" model, where the command center is located in a secluded spot (TechRxiv, 2025).

RELATED STUDIES

Local Studies

The Philippine market is quite welcoming towards the concept of smart homes. Predictions suggest that by 2025, 61.7% of Philippine households will own at least one smart home device, primarily due to the increasing significance of home security and energy-saving measures (Vista Estates, n.d.). This increasing adoption of technology indicates a favorable setting for the implementation of cutting-edge, wireless, and inexpensive intrusion detection systems (Ken Research, n.d.). The high demand for IoT devices with an emphasis on security indicates a promising market feasibility of products such as the Hybrid Home Security System.

Besides academic prototypes, modern-day trends indicate an increasing readiness of the Philippine market towards adopting smart home systems. As reported by Manila Bulletin (2025), adoption rates of smart homes in the country have risen dramatically, with a number of residential developments incorporating advanced features such as fingerprint lock doors, audio and visual intercoms, and intelligent sockets. These are described as convenient, efficient in terms of energy savings, but most importantly, safe systems that are becoming increasingly popular among Filipino homeowners who have become more inclined towards technology-based home solutions as costs fall and supply expands. That is where the Hybrid Home Security System comes into play by providing the system for real-time intrusion detection, with homeowners being notified instantly at a much lower cost compared to commercially available smart security systems. This system not only aligns with technological trends but also directly addresses a gap in the local market, making it both timely and practical

In a local study conducted by researchers at the Philippine E-Journals (2024), a framework was established for a wireless security system using a Raspberry Pi coupled with GSM modules. The research was carried out on the latency of alerts, the time it took to transmit the picture of an intruder's face to the mobile device of the house owner. It emerged that, although the GSM network provided broad coverage, it had low bandwidth, which made the transfer of images difficult.

Other related prototypes from the Philippines would include the study by Balahay, Petiluna, Ponay, Taboso, and Tangkion (2024) entitled "Smart Home Security System Using Raspberry Pi and Arduino Uno With Emergency Response." The researchers utilized the power of two microcontrollers to come up with an advanced yet layered system that is capable of handling devices while incorporating sensors. One of the notable aspects of the system proposed by these researchers would be the inclusion of the emergency response feature, wherein the system could alert users about a possible intrusion. This study will prove useful to our Hybrid Home Security System because it will give us a glimpse into local innovation, although there remains a need to keep the system affordable.

While early local models used only one detection system, recent studies show that a “Hybrid” model is better at avoiding false alarms. According to Gomez et al., "Residential Smart Security in The Philippines" (2025), the combination of sensors and vision through AI is critical to enhance protection.

With regard to the development of the System, the mentioned hybrid architecture will be used to combine the Contact Detection Module at the entrance part with Facial Recognition through the use of the Raspberry Pi 5. Through this combination, the System is guaranteed that it will not only detect the presence of the user but more importantly, the contact made through the opening of the door. With this, the System will conform to the prevailing trend in academic circles within the locality wherein motion sensing has already been replaced with actual detection technology.

A technical analysis done recently by Turingan and Mendoza (2025) analyzed how well the AI surveillance performs in the Philippine scenario. It was discovered that most of the inexpensive facial recognition systems did not work properly because of "environmental noises," such as inadequate light in the corridors and high moisture, which leads to early deterioration of sensors. The results from their study revealed that for an AI surveillance system to function effectively in the Philippine scenario, it needs to use a high-resolution camera along with HDR and Edge AI processors.

Foreign Studies

The key to securing the house lies in understanding the steps that burglars take during an attempt to break into the property. This was evidenced by Kuhns et al.'s (2022) research conducted on the Department of Criminal Justice and Criminology, University of North Carolina at Charlotte, whereby their findings were based on behaviors displayed by intruders who always follow a pattern when planning to invade a certain house. First, they will knock on the front door of the house to ascertain whether anyone is home. If there is no response, the second thing they will do is attempt to unlock the door or windows. And if the intruder is unable to unlock they use physical force.

According to a major study presented in MDPI Sensors (2025), the trend towards home security worldwide has moved towards "Decentralized Deployment." In their study, the researchers proposed a hybrid approach that could work fully on small IoT devices such as the Raspberry Pi. The average accuracy rate was over 95% across varied light settings. Furthermore, another thorough review of IDS presented in the International Journal of Electronics Engineering (2026) shows how IDS have evolved over time. It shows that a single sensor approach (either a PIR alone or a camera alone) is not enough anymore because they can generate false alarms. The researchers recommend "Multimodal Detection" that incorporates inputs from more than one type of sensor, such as contact sensors and visual AI.

In addition to these, some other studies have also examined the performance of PIR motion sensors as the main detector in intrusion detection. For example, in their research, Kazi et al. (2023) designed a PIR security system for Arduino that could detect human motion successfully but also pointed out its incapacity to monitor static openings such as doors and windows. In a similar vein, Akinwumi et al. (2021) designed a PIR-GSM detector that could trigger alarms and send messages when movement was detected. Despite being very effective, the system had the problem of not differentiating between intruders and residents and not having visual feedback in real-time. This issue was solved in the study by Arinde and Idowu (2024), where the authors incorporated PIR and doormagnetic sensors with a GSM module, thus increasing their accuracy.

In today's home security systems, there needs to be a mechanism that would facilitate human presence identification in order to trigger the facial recognition AI. The research undertaken by Nagaraj et al. (2026) illustrates the potential application of the You Only Look Once algorithm in creating an efficient and smart home environment that combines IoT devices. With the ability of the YOLO algorithm to detect and determine the presence of human beings, lights and thermostats in different rooms are intelligently turned on and off without any kind of manual intervention. This was demonstrated in the experiment conducted on the basis of the human image datasets, which proved that the system can accurately classify presence.

Whereas most other international systems concentrate on passive surveillance (alerts), it indicates a return to "Proactive Protection." Their study revealed that alerts confirmed by AI and audible have proven far more efficient in deterring burglary than passive alerts. It explains how an immediate response with audible signals compels an "instant flight" in intruders, whereas silent notifications often allow the intruder enough time to finish the crime before the owner or police arrive.

Integrating IoT platforms with mobile applications has been considered very important for usability and responsiveness. According to a research study conducted by

Universitas Sumatera Utara (2022), a home security system based on ESP32 technology was created using the Blynk app, resulting in an average response time of below two seconds and illustrating the ease of use of mobile integration. In another case, a smart home system was proposed by STMIK Kaputama (2023) that integrated PIR sensors, CCTV cameras, and environmental sensors with ESP32 microcontrollers controlled via a mobile application.

On a larger scale, Torres-Hernandez et al. (2025) did a meta-study by examining over 40 cases of IoT home security systems. They discovered that homeowners were more satisfied and felt safer when they had real-time notification systems than those who just had CCTV or alarm systems. This sociotechnical approach explains why Hybrid Home Security Systems focuses on real-time notifications to enhance both its performance and users' satisfaction.

IoT security systems have been designed to cater to unique scenarios outside of domestic settings as well. In one such case, Universitas Pertahanan Republik Indonesia (2024) has designed a surveillance model using a PIR motion sensor coupled with ESP-32 WROVER-CAM to be used in military establishments and other highly secured locations. The system was specifically designed to detect motion and immediately transmit live video feeds to authorized personnel through IoT-enabled communication platforms. The researchers pointed out how the use of cheap sensors combined with microcontrollers would make it possible to achieve an equal level of performance to commercial surveillance camera systems, albeit much cheaper. The researchers also pointed out how the ability to access videos in real time would enable security teams to verify potential threats immediately, mitigating any dangers posed by delays in response or false alerts.

This concept of being affordable, reliable, and adaptable can be applied within domestic settings

In addition to this, there is an analysis of IoT-based security systems, with an emphasis on the system's architecture, functionality, and effectiveness within smart homes. From their research, it can be observed that the inclusion of sensors into wireless communication systems (such as Wi-Fi or GSM) was the key factor that contributed towards the greater flexibility and ease of use of IoT-based systems in comparison to wired alarm systems. Moreover, it has been stated that PIR sensors continue to be the most widely used sensors for the detection of movements because of their cost-effectiveness and reliability; however, the effectiveness of these sensors is improved when used alongside other sensors, such as magnetic sensors or vibration sensors. Moreover, mobile applications were found to play an important role in IoT-based security systems by allowing homeowners to have better control and monitoring capabilities over their home security systems.

The speed of sending an alarm in home security is extremely crucial. Based on research from Universitas Sumatera Utara (2022), the security system that uses an ESP32 microcontroller can detect the intrusion and deliver alerts within less than 3 seconds, with a recorded average of just 1.8 seconds. This quick process is essential because it will instantly notify the homeowner before anything dangerous occurs.

Conceptual Framework

The framework depicts the study's broad outline. It can be phrased as follows: the input-process-output (IPO) framework.

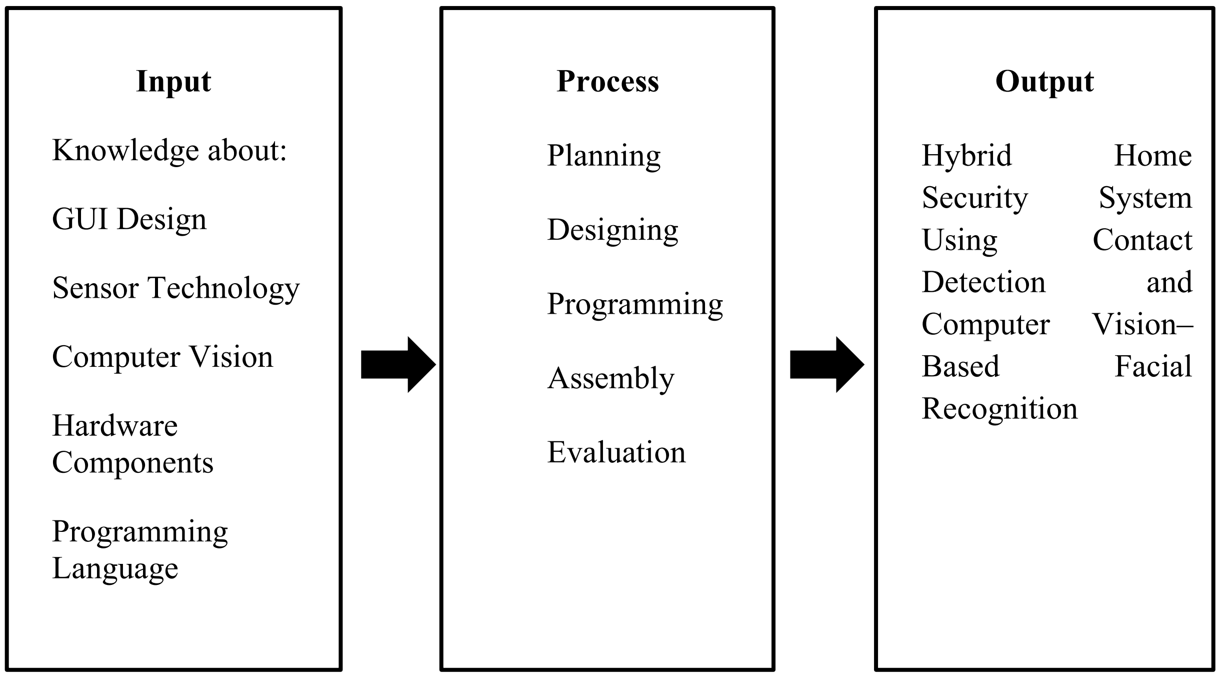

Figure 1. Conceptual Framework

The input part covers all of the knowledge and abilities needed to develop the a hybrid home security system. It involves understanding GUI design in order to create the system interface, sensor technology for detecting physical intrusion, computer vision for implementing facial recognition, programming languages for developing the system's software, and hardware components necessary for building the physical device. Such inputs provide the developers with all that is necessary in terms of means and information in order to build a security system.

The Process explains the order of operations which will be used to convert inputs into the final product. In the planning phase, project objectives and system requirements are validated. Further on, during the Designing phase, the system structure, workflow, and hardware configuration are described. During programming, all required software functions for sensors' operation, system control, and alarm activation will be developed. The assembly phase deals with hardware integration and linking necessary for the system to operate. Finally, in the evaluation phase, the whole system is tested thoroughly in order to fully assess its efficiency.

The Output block is the final outcome, a ready-to-use Hybrid Home Security System Using Contact Detection Module and Computer Vision-Based Facial Recognition. The system is capable of detecting contact-based intrusions and identifying authorized residents from unauthorized individuals through its facial recognition feature, which produces necessary alerts through its local mechanisms for reliable home protection.

Chapter III

METHODOLOGY

This chapter presents the methods and procedures that will be used in conducting the study. It describes the research locale, respondents, research design, research instruments, procedures, and data analysis techniques that will guide the development and evaluation of the system.

Research Locale

The study will be conducted in one of the researchers' houses in Lucena City, Quezon Province. This is a busy city area that provides the right environment for running and testing the security system. The testing and evaluation will be conducted in a home. setting, particularly on the first floor of residences in Lucena City, where common security issues typically arise.

Respondents

Respondents in this research will be the researchers themselves. Because of the need to control the tests conducted on the home security system, the researchers will conduct a preliminary test to determine how the system works and how effective it is in detecting intruders in the home. As respondents, the researchers can test the system throughout their research process and discover problems that might be inconvenient for homeowners. This approach ensures that the system is effective and efficient for home security.

Research Design

This study will use Applied Research Design. The purpose of this design is to create, develop, and test a system that addresses a specific problem, in this case, the lack of immediate awareness in home security. The Hybrid Home Security System will be developed with a contact detection module, implementation of a facial recognition system using computer vision algorithms, together with a sound alert. The developmental part of the design focused on the step-by-step process of building the system: planning, designing, programming, assembling hardware components, and evaluating the system. The researchers chose this research design because it not only allowed the researchers to develop a working system but also to test and evaluate how effective and efficient it is in solving the problem of delayed awareness in home security

Research Instrument

To gather relevant information for this research study, data were collected through the following sources:

1. **Internet**

   Through the internet, the researchers review related literature and studies that are connected to this study, which will help in creating and developing this study. Also the internet helps to obtain information about the materials that can be possibly used in this study.

2. **Published Materials**

   The published materials, such as electronic books and journals, were also used for the related literature and studies that are the most credible resources for the study. The researchers also used this as a source to know how the materials would be used in the study.

3. **Academic Databases**

   These are the IEEE Xplore, ResearchGate, and Google Scholar, which give credible resources related to Home Security Systems.

Procedures/Data Collection

Planning

In this phase, the researchers will identify the goals, requirements, and scope of the home security system. They will determine the hardware components needed, such as sensors, a camera, and a speaker, as well as the software tools and computer vision resources required. The research team will create the entire design of the Hybrid Home Security System process. That means that the researchers will decide how it will be possible to recognize faces and determine if the person is among the users. The process of face identification is ready to be applied efficiently. It gives a guideline for the further development of the system.

Designing

In the design stage, the design of the structure of the system will be made by the researchers. They will decide on how to connect all the components in relation to the Contact Detection Module, Camera Module and even the Raspberry Pi 5 with a 7-inch touchscreen. They will also discuss here the designs of the casing of each of the components to make sure that the measurement is appropriate. In this part, they will show the design of the Graphical User Interface of the Hybrid Home Security System.

Programming

The programming stage is where the actual code for the system will be written. This includes programming the microcontroller to detect the presence of the contact sensor, programming the algorithm for face detection, and programming the speaker to speak upon an alarm.

Assembly

In the assembly, the following hardware components will be brought together by the researchers. These are connecting the contact sensors with the microcontroller, installing the camera for facial recognition, and the speaker for sound alerts. All components will be assembled in such a way that they form one complete system.

Evaluation

During the evaluation stage, the complete system is tested for its performance. The investigators will test the effectiveness and efficiency of the contact detection system for detecting interactions and the face recognition system for identifying people, and whether the sound alert is triggered effectively or not. Any issues discovered during testing will be corrected to improve the system's effectiveness and efficiency.

Chapter IV

RESULTS AND DISCUSSION

This chapter discusses the data that was gathered from the development and testing of the system. It contains circuit designs, program codes, flowcharts, and a list of all the components that were used. It also covers the analysis and discussions of the collected data to show the performance of the system.

System Overview

The Hybrid Home Security System is an automated, security tool designed for break-ins and theft by alerting homeowners of any unauthorized access that is detected through entry points of the house. The system is built using a Raspberry Pi 5 equipped with an Active Cooler and 27W Power Supply which acts as the main brain, and several Raspberry Pi Zero 2 W units that serve as remote wireless cameras for different rooms.

The System uses a Wireless Contact Detection Module. This module is built with an ESP32 Super Mini that transmits information from Magnetic Reed Switches and MPU6050 sensor to Raspberry Pi 5 when an unrecognized activity is detected. These modules are powered by Lithium-ion Batteries, and a MAX17043 fuel gauge chip sends the battery percentage to the 7-inch Raspberry Pi Touchscreen so that the user knows when to charge them. The contact detection module detects activities like opening, knocking, wiggling, or forcing the entry points.

Using custom Python code, the system checks the camera image to see if the person is one of the registered users or not. If the camera recognizes the person as an authorized user, the sound alert stays inactive even if they touch or open the door. if the camera detects an unrecognized face, any contact with the door will immediately trigger a loud warning sound alert through an 8-ohm 3W speaker. Everything is controlled through a 7-inch Touchscreen, making it easy for the homeowner to see the status of the whole house.

Project Technical Description

This section shows the preparation for the creation of Hybrid Home Security System.

Planning

The planning section presents the initial preparation and technical basis for developing the Hybrid Home Security System. It discusses the system's core components, selected sensors, camera options, facial recognition process, block diagram and flowcharts,. This section serves as the system processes needed to achieve the intended security functions.

Table 1 presents the 3 main intruder interactions, like knocking, wiggling and forcing, that the system's contact detection module will be designed to recognize. Before the intruder breaks into a house, they usually follow steps to check if it is safe to enter the house. The table explains the meaning of each action, the intruder's goal for doing it, and why it is needed for the security system to detect those interactions.

Table 1. Identified Intruder Interactions for Systems Contact Detection

Intruder Interaction	Description	Intruders Purpose	Purpose of Detection
Knocking	Tapping on the door or window	To check if someone is inside the house before attempting to break in	To provide early warning and detect the intruder before they even try to open the door.
Wiggling	Testing doorknobs, shaking the window or picking locks	To check if a door or window is unlocked	To detect unauthorized tampering or someone trying to silently enter the house
Forcing	Kicking the door or breaking the windows	To gain entry using physical force	To immediately trigger the alert and call for help.

Table 1 shows the identified interactions that represent the actions in the burglar's attempt to enter. Knocking is usually the first interaction intruders use to check if the house is empty. If the house is empty, the intruder proceeds to wiggle the doorknobs or window frame to unlock an easy entry point. Detecting wiggling interaction prevents the silent intrusions. Finally, detecting the forcing interaction allows the system to immediately trigger the main alert and call for help. Overall, detecting those 3 interactions ensures that the system can protect the home and notify the homeowner of the burglar attempt.

Components of the Hybrid Home Security System are shown in Table 2. The table indicates the main components of the hybrid home security system, along with their function and how they help in the detection of any door/window movement, facial image acquisition, recognition of registered users, triggering of alerts, and control of the entire system. Understanding the components of the hybrid home security system helps in comprehending how these components perform security-related functions.

Table 2. System Core Components/Module Functions

System Components/Module	Description	Function
Contact Detection Module	Uses a sensor for the entry door/window.	Capable of detecting door/window interaction.
Camera Module	Captures a person's face at the entry point.	Provides a clear image input for the facial recognition process.
Facial Recognition System	The process of comparing captured faces with the registered ones.	Determines whether the detected person is a registered user or an intruder.
Sound Alert System	Uses a speaker to warn the users about unrecognized activity.	To produce an audible alert when an unrecognized activity is detected
Control and Monitoring Hub	Provides a system processing and monitoring interface for the security system.	Controls system operations, processes sensor and camera data, and displays monitoring information.

The module of the proposed Hybrid Home Security System has been discussed in Table 2. The table contains the names of the different modules and functions performed by them, which together form the basis of the security system for a house. The Contact Detection Module monitors contacts from any person on the doors and windows using sensors. Camera Module clicks pictures of people’s faces at the pre-entry and entry points of doors to facilitate monitoring and identification. The Facial Recognition System compares the picture taken at entry with the faces of registered persons already existing in the system to verify whether the face being detected is registered or not. Sound Alert System produces an audible sound alert when an unauthorized person enters or when an unauthorized person is detected at doors and windows. The Control and Monitoring Hub acts as the controlling part of the system which controls the functioning of all modules and provides useful information to the user.

Table 3 compares the listed cameras based on key technical specifications that may affect the performance of the facial recognition module of the Hybrid Home Security System. These cameras are used to capture images of registered and unregistered users at entry points for authorized access verification. The researchers compared camera resolution, field of view, video transmission speed, low-light performance, connectivity, and compatibility with computer vision processing.

Table 3. Specification-Based Comparison of Cameras for the Hybrid Home Security System

Specification	Raspberry Pi Camera Module 3	IP Camera	Analog CCTV Camera
Resolution	12 Megapixels	2 Megapixels (1080p) to 8 Megapixels (4k) depending on model	1 Megapixels (720p) to 2 Megapixels (1080p)
Field of View	75° (Standard version) or 120° (Wide angle version)	90° to 130° (varies by manufacturer)	60° to 90° (Standard CCTV viewing angle)
Video Transmission Speed	Up to 50 - 120 FPS via direct connection	15 - 30 FPS via wireless/wired network connection	25 to 30 FPS via analog transmission
Lens / Focus	Phase Detection Autofocus, standard and wide-angle variants	Fixed lens or varifocal lens depending on model	Usually fixed lens, some models have manual varifocal lens
Night Vision Capability	Specific "NoiR" model and external Infrared (IR) light	Equipped with built in Infrared (IR) LEDs	Equipped with built in Infrared (IR) LEDs
Connectivity	CSI-2 ribbon cable directly connected to Raspberry Pi	Ethernet, Wi-Fi, RTSP/ONVIF depending on model	Coaxial/BNC cable connected to DVR
Computer Vision Compatibility	Direct access to image frames using Raspberry Pi libraries	Requires network stream configuration and video decoding	Requires DVR, capture card, or converter before image processing
Main Limitation	Short ribbon cable and needs Raspberry Pi hardware	Possible network delay and configuration complexity	Not directly compatible with real-time computer vision processing

Table 3 shows the specifications on how the researchers will choose the appropriate camera for the study. The 3 cameras that are being compared have different strengths and weaknesses, but the Raspberry Pi Camera Module 3 stands out as the best choice for the facial recognition of the system. It takes pictures at 12 megapixels, and it can take videos really fast, up to 120 FPS. It also has autofocus, which helps it take pictures of moving faces.

The other cameras, like IP cameras and analog cameras, are not as good as the camera module 3. Even though the IP camera and analog CCTV can see at night, the picture is not very clear. IP cameras can delay if the WiFi is slow. An analog camera needs extra hardware that converts analog to digital so that it can work with a computer.

The Raspberry Pi Camera Module 3 is still the choice. It is the most compatible camera for studying. Because it can recognize people with low delay, the Raspberry Pi Camera Module 3 is a great choice for security systems. The researchers identify the core components of the system.

The comparison of the sensors that have been proposed for the Contact Detection Module of the Hybrid Home Security System is presented in Table 4 below. In this table, the roles and applications of the sensors are discussed. The interaction types that can be detected by the sensors and their advantages and disadvantages are also explained. In order to select sensor components suitable for detecting events related to doors and windows such as being opened/closed, manipulation attempts, and burglary, this comparison was conducted by the researchers. From this comparison, the most reliable and accurate sensors for contact detection were identified to enhance the security features of the system.

Table 4. Comparison of Sensors for the Contact Detection Module

Component	Purpose/ Function	Sensor possible interactions	Advantages	Limitation
Magnetic Reed Switch	Detects opening and closing interaction.	Open and close	Reliable detection of door/window opening and closing	Cannot detect vibration, interaction
Piezo Electric	Detects vibration and physical impact	Vibration and impact	Sensitive to strong impacts and force	Less accurate for detailed motion detection
MPU6050	Detects motion, tilt, and orientation.	Motion, tilt and shaking	Detects motion tilt, and tampering with higher accuracy	Requires calibration for accurate detection
KY-038	Detects sound intensity or loud noises.	Sounds	Detects unusual loud sounds near the monitored area	Sensitive to environmental and background noise

Table 4 shows a comparison between the sensors that have been evaluated in the design of the Contact Detection Module of the Hybrid Home Security System. Each sensor has been compared on the basis of its sensing properties, interaction parameters, merits, and demerits. On the basis of the above comparison, Magnetic Reed Switch and MPU6050 have been found to be the most suitable hardware devices. The Magnetic Reed Switch is preferred for providing accurate and definitive confirmation of whether doors and windows are open or closed.

Moreover, the MPU6050 sensor was preferred over the Piezoelectric and KY-038 Sound sensors due to the continuous multivariable telemetry feature provided by this sensor. Whereas piezoelectric and sound sensors detect only generic physical shock and loud ambient noise, respectively, the MPU6050 sensor incorporates both an accelerometer and a gyroscope that help accurately measure any movement or shaking. More importantly, through mathematical calculation of sensitivity limits, the MPU6050 sensor makes it possible to distinguish natural vibrations and physical shock from malicious activity. Therefore, the use of both the Magnetic Reed Switch and the MPU6050 sensor is highly reliable for this application.

Table 5 shows the design considerations of the Contact Detection Module of the Hybrid Home Security System. It indicates the significant elements that were taken into account when developing the module, including ease of installation, continuous operation during brownouts, wireless installation, and quick alert transmission. The table also shows the corresponding actions taken by the researchers.

Table 5. Design Considerations of Constructing the Contact Detection Module

Consideration	**Description**	**Action**
Easy installation and continuous operation during brownouts	Independent device. Can be placed anywhere in the door or window without the need for a wall plug. And keeps working during brownouts.	Use a rechargeable battery
Clean Setup	Wireless Connection. Can be moved without worrying about cables.	Use a wireless sensor (ESP32)
Fast Alert and Lightweight	Instant Alert and Efficient. The device should instantly give the data to Pi 5 the moment the door or window is triggered.	Use an MQTT

Table 5 presents that the Contact Detection Module is built to work independently. By using a rechargeable battery, the device can be placed on any door or window without needing a wall outlet. And it will keep working even in brownouts. Also, using an ESP32 as a wireless microcontroller keeps the whole setup clean and removes any messy wires and prevents tripping. Lastly, the implementation of the MQTT protocol provides a lightweight and efficient communication link, which ensures that the Raspberry Pi 5 is notified when a sensor is triggered.

Table 6 presents the planned process of the Facial Recognition System for the Hybrid Home Security System. It shows the purpose of each process and how human detection, face detection, and face recognition will be performed.

Table 6. Plan on the Facial Recognition System Process

Process	**Purpose**	**Computer Vision Algorithm Used / Component Used**
Human Detection	Detects the presence of a person within the monitored area before facial processing begins.	Day Light: YOLO Low Light: PIR + LED + YOLO
Face Detection	Detects where the face is located the video frame from Yolo.	SCRFD
Face recognition	Verifies whose face is detected.	ArcFace

Table 6 presents the planned process of the Facial Recognition System for the Hybrid Home Security System. The system begins with Human Detection, where the presence of a person is identified within the monitored area using the YOLO algorithm. In low-light environments, the system utilizes a PIR sensor and LED lighting together with YOLO to improve human detection capability.After the detection of a face, Face Detection will use the SCRFD algorithm in order to pinpoint the location of the face in the video frame. Finally, Face Recognition will use the ArcFace algorithm in order to verify the face against the faces in the database. These processes work together to provide accurate identification and monitoring of authorized and unauthorized individuals within the secured area.

Block Diagram

The diagram below illustrates the visual representation of the system and its functionality. This represents the connections between all the different parts of the system, including the inputs, process, and output.

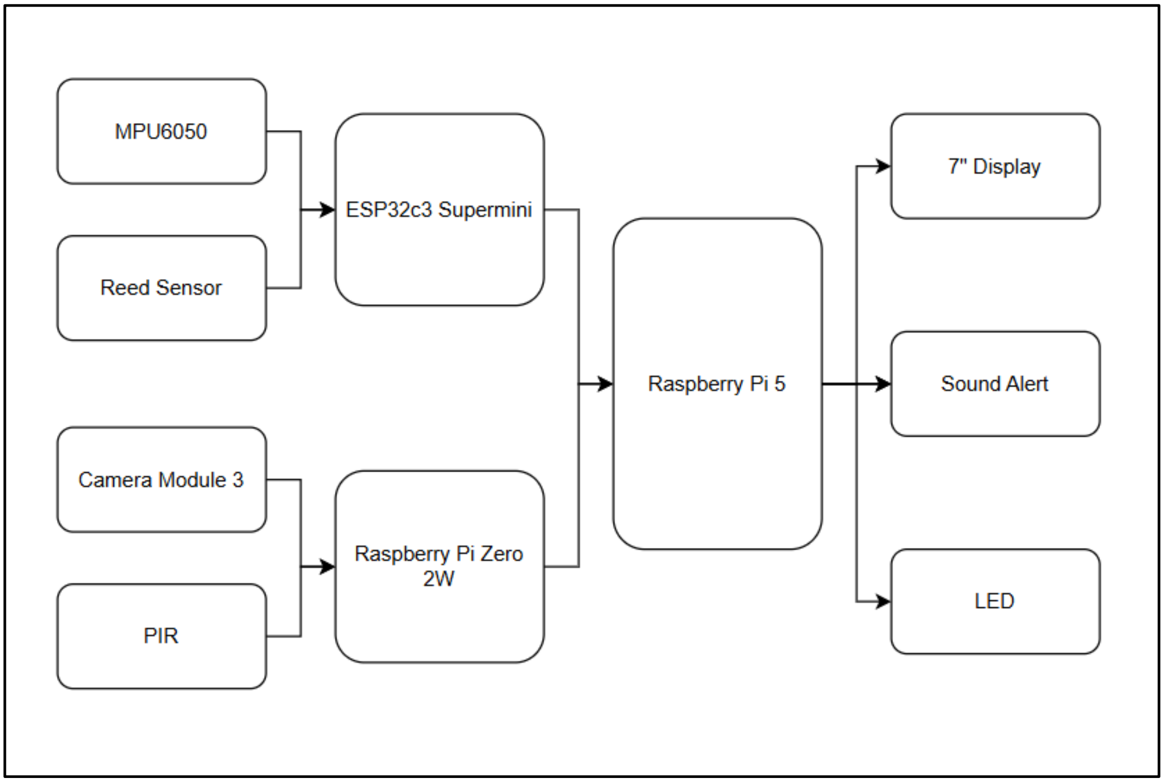

Figure 2. Block Diagram of the Hybrid Home Security System

Figure 2 displays a block diagram illustrating the input, process, and output in the system, providing a visual representation of the process involved for easier understanding. The block diagram begins with the input, which includes the MPU6050 and Reed Sensor, sending data to the ESP32c3 Supermini, while the Camera Module and PIR sensor connect to the Raspberry Pi Zero 2W. These two microcontrollers send their information to the Raspberry Pi 5, which acts as the main brain of the system. Lastly, the Raspberry Pi 5 triggers the outputs on the right, which are the 7-inch display, the sound alert and the LED light that notifies the user of any activity.

Flowchart

This section presents the flowcharts for the Hybrid Home Security System. This process guides the user on how the Camera Module for facial recognition and the Contact Detection Module work.

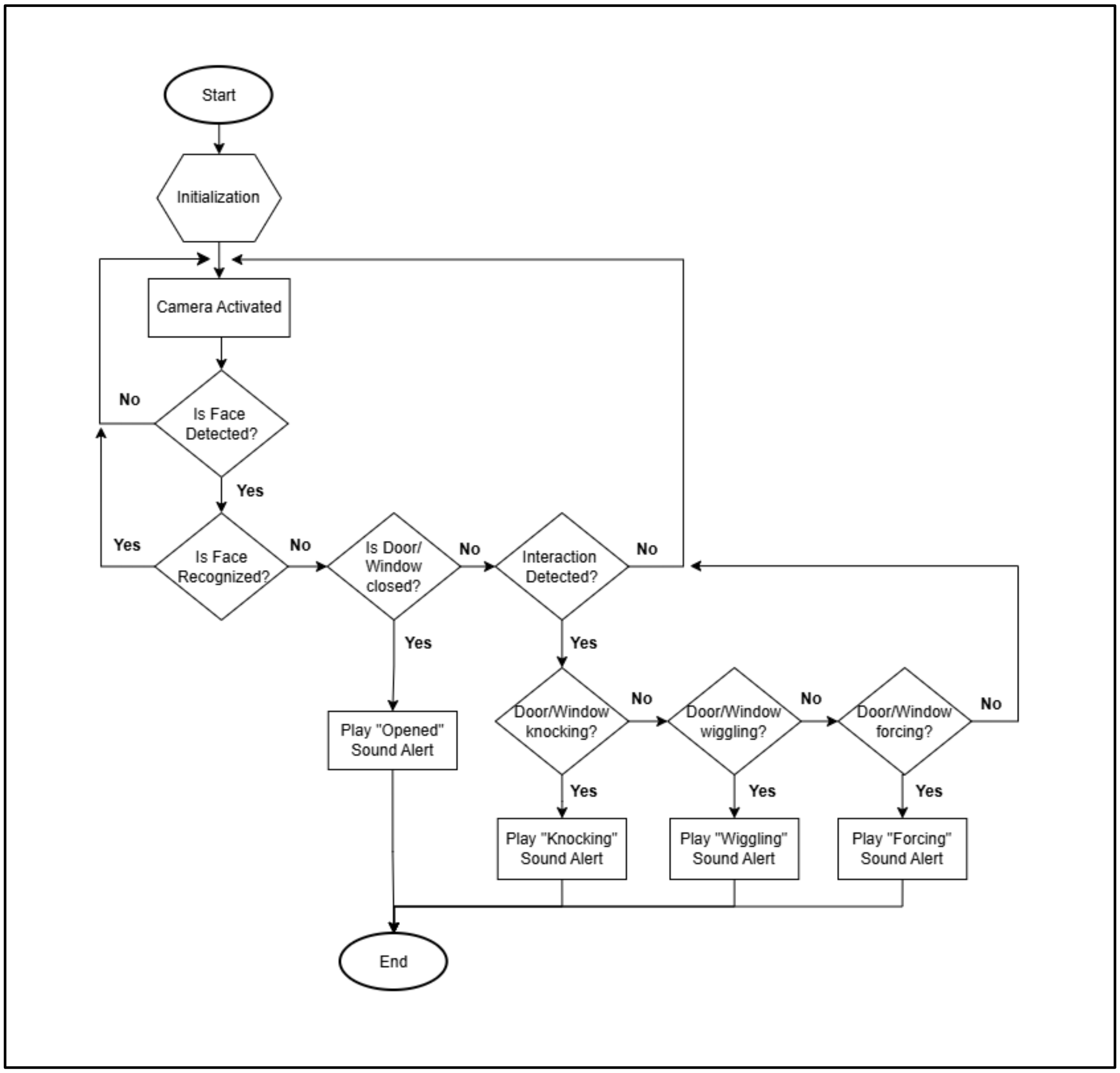

Figure 3. Hybrid Home Security System Flowchart

Figure 3 illustrates the flowchart for the Hybrid Home Security System. The process initiates with a system start and initialization phase, which immediately activates the integrated camera. The system enters a continuous monitoring loop, first scanning the visual feed to determine if a human face is detected. The absence of the face causes the module to go back to maintaining surveillance through the camera. If a face is identified, then the next action involves the identification of whether it is a face of an authorized user. If the face is known, the system goes back to the camera activation phase. But if the face is not identified, the system will proceed with the use of sensors in determining whether there are threats. The first check will be on the state of the door or the window; if it is found to be open, an alert "Opened" sound will be generated immediately. But if the door or window remains closed, then the sensor detects physical interactions. On detecting interactions, the module will classify them in three levels: knocking, wiggling, or forcing. Based on the intensity, appropriate sounds will be triggered accordingly. When there are no interactions, the module goes back to the camera activation phase.

Designing

The designing part describes the suggested configuration and design of the Hybrid Home Security System. It provides the details of the circuit design, case design, position of sensors, GUI design, and software/hardware implementations that help in the development of contact detection, camera, and monitoring applications. These designs provide the foundation for the proper functionality and appropriate design for home security use.

Circuit Plan

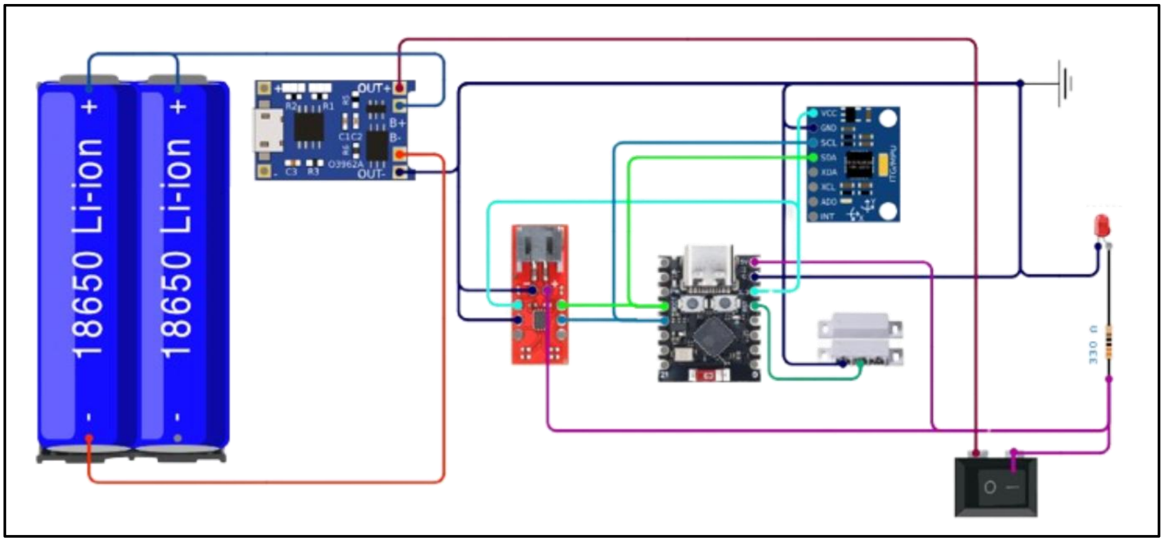

Figure 4 . Circuit Plan for Contact Detection Module

The Circuit Diagram of the Contact Detection Module is shown in Figure 4 below. Both the Magnetic Reed Switches and MPU6050 Sensors are attached to the ESP32 Super Mini for detecting whether a door is open or whether there is any interaction with it. A red LED and a resistor are used to show that the system is active. The module is powered by a lithium-ion battery, which allows it to work without being plugged into a wall. This battery is managed by a Type-C charging module for easy recharging and a MAX17043 battery fuel gauge IC to monitor the battery level. A rocker switch is also included to turn the power on or off. When the sensors are triggered, the ESP32 sends wireless data to the Raspberry Pi 5. This makes the module portable, rechargeable, and easy to use anywhere in the house

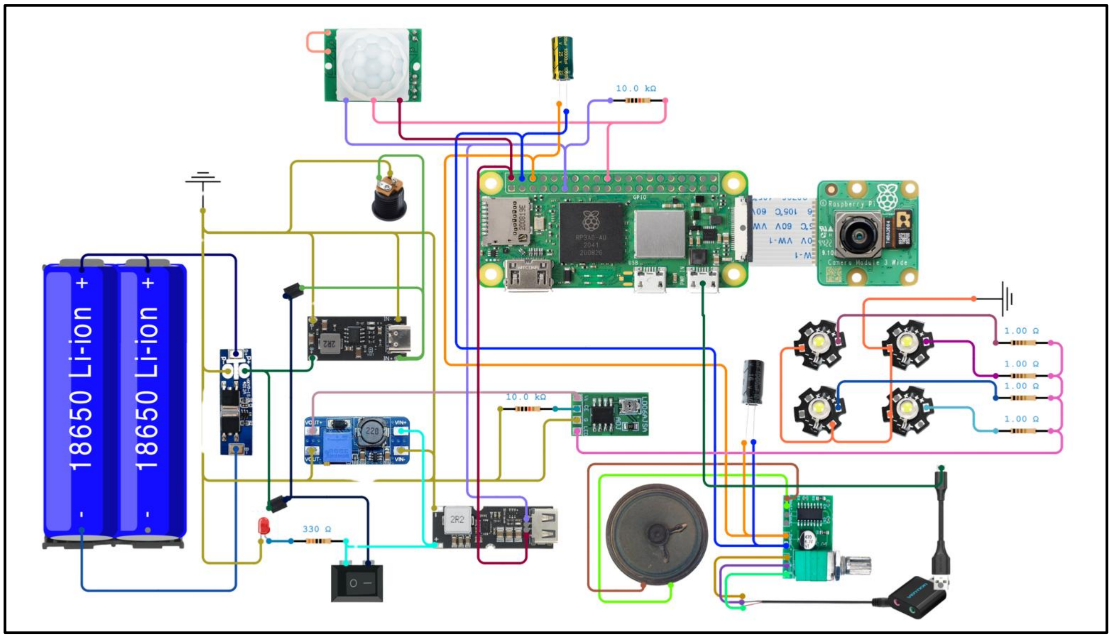

Figure 5. Circuit Plan for Camera Module

Figure 5 shows the circuit plan for the Camera Module. This module uses a Raspberry Pi Zero 2W as the main controller and is connected to the Camera Module 3 for live feed. A PIR HC-SR501 motion sensor is also connected to detect movement. This Camera Module uses electricity as the main power, but also has a rechargeable battery as a backup when the power suddenly cuts off. The rechargeable batteries are connected to regulator boards to ensure that the camera still functions and the electricity is steady and safe, so that the components don't get damaged. Also, this includes an LED bead surrounding the camera to provide lights in low-light surroundings. This has an 8-ohm 3-watt speaker connected to an amplifier, which is the PAM8403, to play sound alerts.

Case Design

The figures below present the case design used to house the Raspberry Pi 5 with a 7-inch touchscreen monitor, a contact detection module and a Raspberry Pi 0, together with a camera module 3 and a speaker.

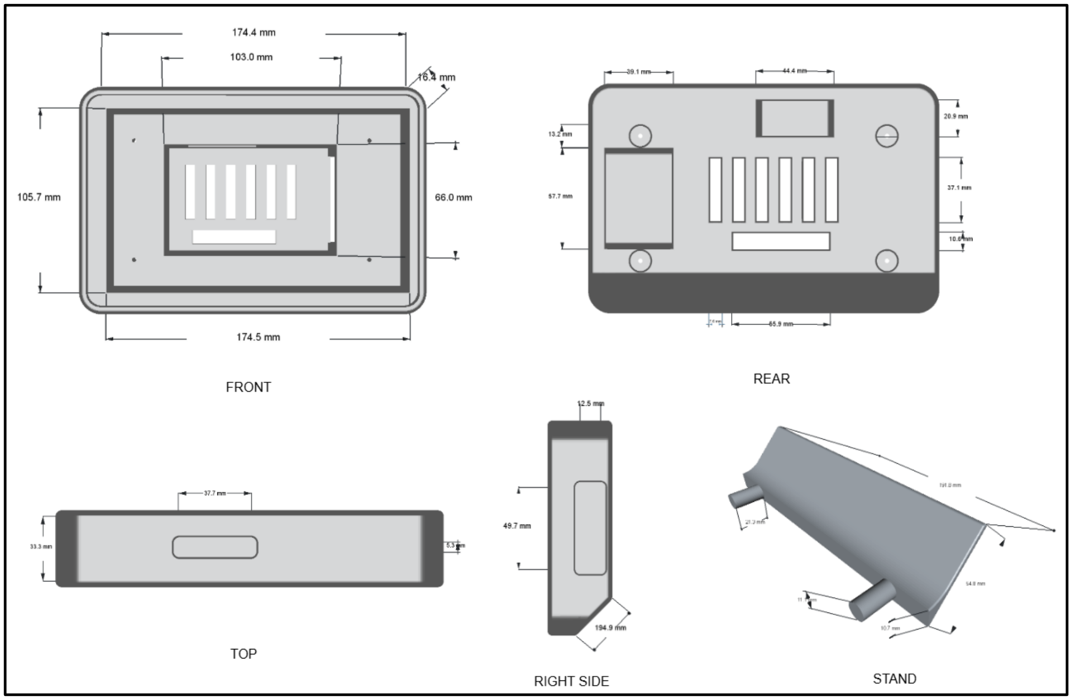

Figure 6. Raspberry Pi 5 with 7-inch touchscreen Case Design

Figure 6 illustrates the full case design enclosure for the main hub, which is specifically sized to fit a 7-inch touchscreen monitor and Raspberry Pi 5. This case shows a precise size of 174.4 mm by 105.7 mm to ensure that the display is centered and is easy to access for touch interaction. Behind the screen is where the Raspberry Pi 5 will be positioned with built-in ventilation slots so that it can manage the heat during operation. On the right side, there's a hole for ports so that the user can easily connect USB connectors whenever they want. As for the stand, it is designed to make the main hub stand in a bit slanted position.

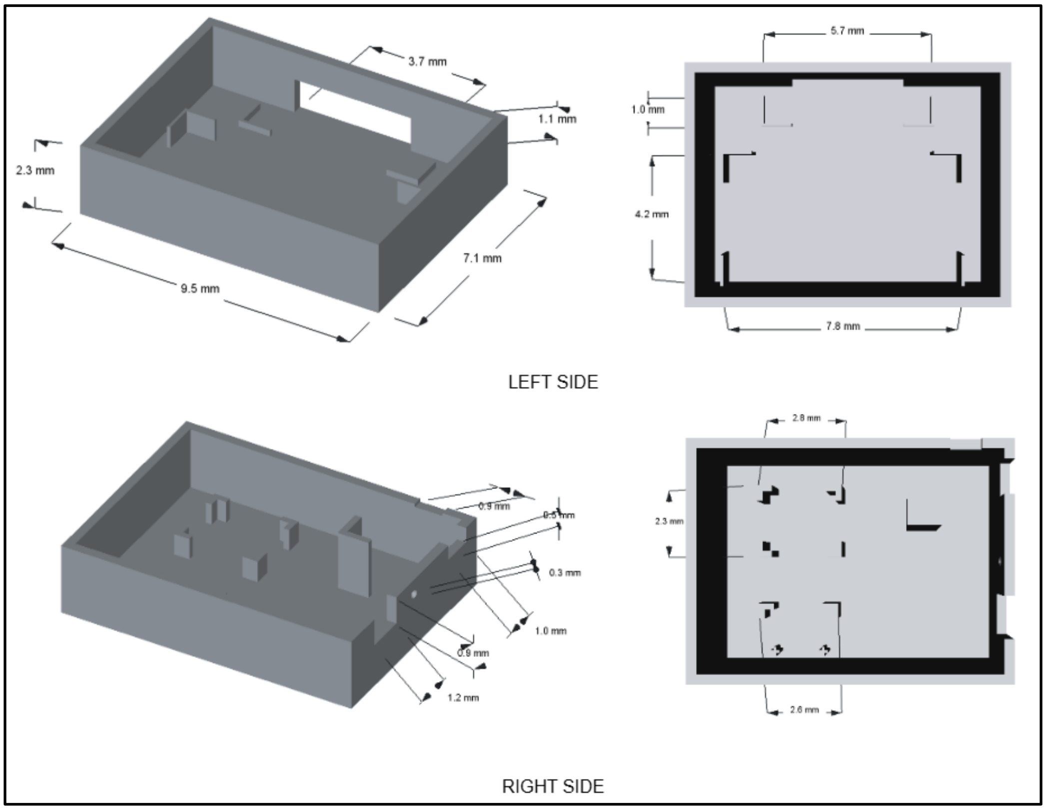

Figure 7. Contact Detection Module Case Design

Figure 7 shows the detailed case design for the Contact Detection Module enclosure which is a key component of the System's security hardware. The case design was split into two views, which are the left side view and the right side view, showing how the internal components are positioned inside. This is a compact rectangular-shaped case with measurements of 9.5 mm by 7.1 mm, with a height of 2.3 mm on each side and 4.6 mm in full height. Inside the case, there are pillars or dividers to hold the components, such as the battery with battery holder, magnetic reed switches, MPU6050 , ESP 32 Super mini and type C charging module in place. On the left side, there's a 3.7 mm opening for the magnetic reed switch so that it can have an external connection to the other side of the magnetic reed switch to properly detect when a door /window is opened or closed. On the other side which is the right side, there's also a cutout of 1.2 mm for the switch and a 0.3 mm circle shape for the LED indicator so that the user knows if the contact detection module is on or off. The 0.5 mm cutout is for the port of the type C charging module, and the 0.9 mm cutout beside it is for the type C charging module indicator.

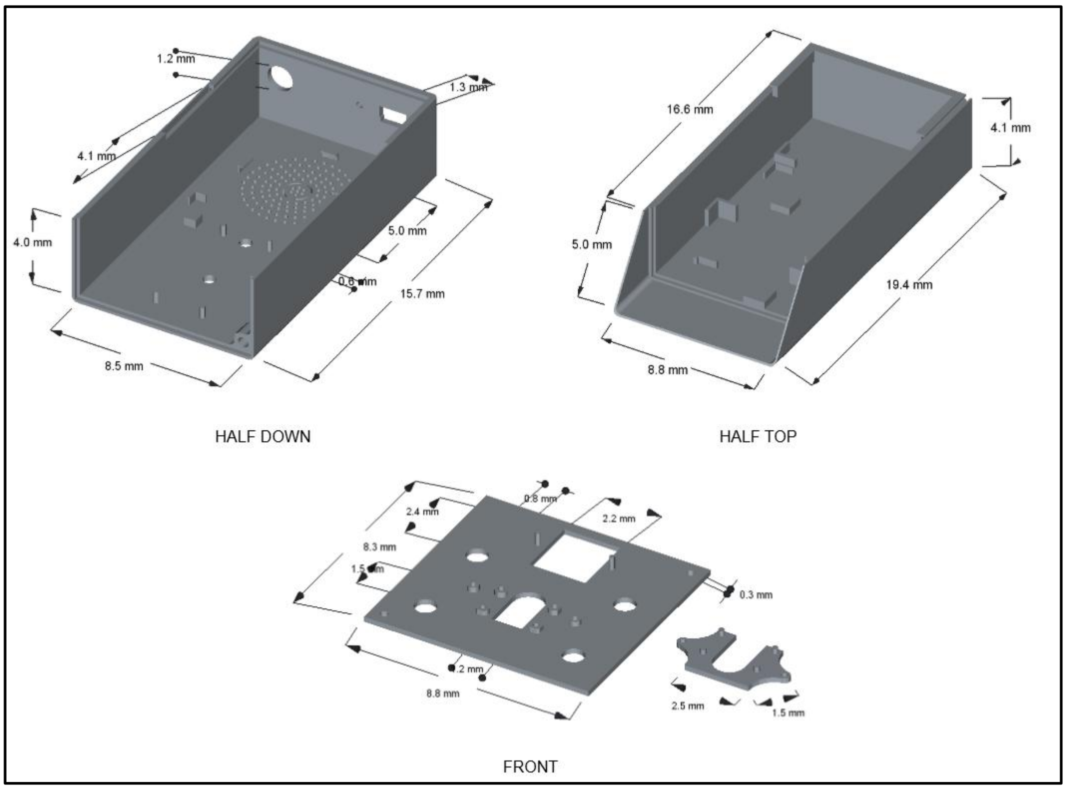

Figure 8. Camera Module Case Design

Figure 8 shows the enclosure case design for the Camera Module. Just like the Contact Detection Module case, this is also split into two sides, which are the half down and half top, together with the view of its front. The half-down section has a measurement of 15.7 mm by 8.5 mm with a height of 4 mm. On the other hand, the half top section has a measurement of 19.4 mm by 8.8 mm with a height of 4.1 mm. Inside of it, both feature an internal pillar or dividers just like the Contact Detection Module case for the firm position of the components. The 1.3 mm measurement is for the switch of the Camera Module, while the 1.2 mm is for the nut of the power adapter. There's also a lot of holes in a circular shape with 5 mm, which is meant for the 8 ohm 3W speaker. As for the half top, its length is greater than the half down since it has a shield roof to protect it from environmental debris. As for the front view, this is where the camera will be positioned and will be locked up so that it has a firm position; it has a measurement of 8.3 mm by 8.8 mm. The 0.8 mm measurement is for the LED bead that will be used when the surroundings are in low light

Positioning of the Contact Detection Module

Figure 9 shows how the contact detection will be placed in real setup.

Figure 9. Positioning of the Contact Detection Module at the Door

Figure 9 shows the placement of the Contact Detection Module installed near the door lock and knob area, together with the magnetic reed sensor attached to the door frame. This module was placed here purposefully, as this is one of the spots where the most attempts at forced entrance take place. Usually, the spot where most of the force or vibration is directed while trying to open a door is around its lock or handle. Thus, the magnetic reed sensor was put between the door and its frame while the detection module was installed close to the knob region, as it would be able to detect vibration, knock, wiggle and force. As stated by Clearway Security (2024), doors, locks, and entry points are often targets for home invasion and break-in due to easy and fast accessibility. Moreover, it is noted that front doors, sliding doors, and windows are particularly susceptible to break-in attempts due to weak locking mechanisms.

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Figure 10 . Positioning of the Contact Detection Module at the Window

The placement of the contact detection module and magnetic reed sensor used in the Hybrid Home Security System is illustrated in Figure 10. This component is fitted in the sliding window along with the magnetic reed sensor which was fixed in the window frame. The contact detection module and the magnetic reed sensor were purposely mounted in such a way that the system would be able to detect the opening and closing movements of the sliding window and detect any attempt at forced entry. The contact detection module was placed closer to the opening part of the sliding window since windows and sliding access doors are usually targeted by criminals who try to intrude into residential houses. According to Homebuilding and Renovating (2025), criminals usually take advantage of weak or vulnerable windows, sliding doors and openings because of their easy access to residential homes. The study further noted that contact sensors and window sensors are more efficient when they are installed close to the frame and opening parts of windows. In addition, the magnetic reed sensor is placed close to and adjacent to the contact detection module so that the sensor can detect the gap between the sliding panel and the window frame during movement.

Graphical User Interface

The figures below show the Graphical User Interface of the Monitoring Application of the Hybrid Home Security System. This is where the user controls and manages the System

Figure 11. Admin Registration Interface

Figure 11 shows the System Initialization when the System is being set up. For the initial set-up, the user must first register the Admin in order for the System to be secured. By clicking the "Scan Admin Face" button, it will capture the face of the person who is in front of the camera. After scanning successfully, a confirmation message will appear to inform the user that the face scanned has been registered as the Admin, and the System is now secured.

Figure 12. Hybrid Home Security System Main Interface

Figure 12 shows the Main Interface of the Hybrid Home Security System Monitoring Application. It displays the menu icon buttons for Detected Intruder, Device Control, Live Feed, Audio Library and Settings. Here, the date and time are also displayed for accurate timestamping of security.

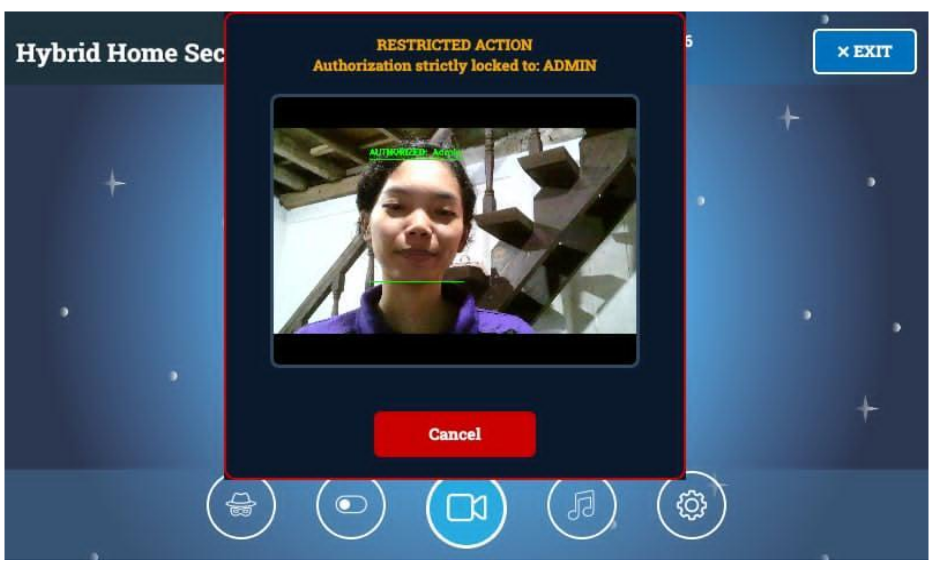


Figure 13. Admin Authorization for Setting Interface

Figure 13 shows the security step of the System when the setting icon button is clicked. Since the setting holds the power in managing the system, an authorization by the admin allows it to be protected from any changes.

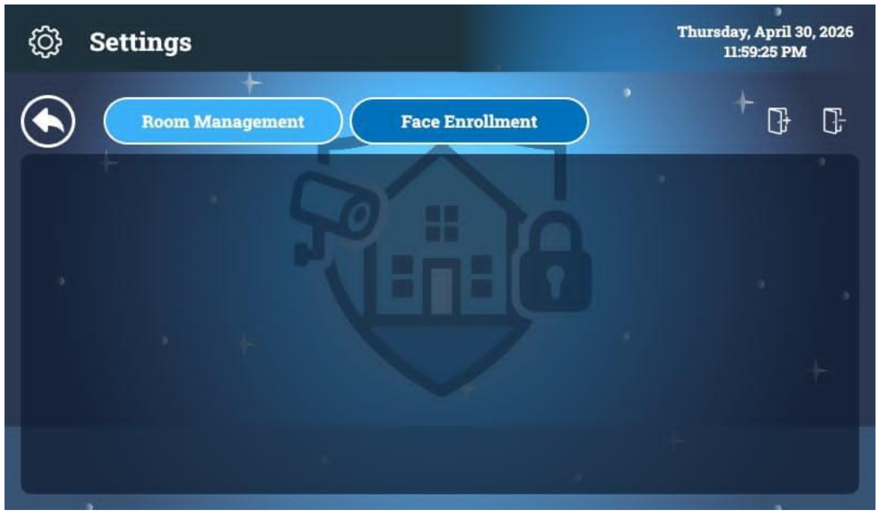


Figure 14. Settings Room Management Interface

Figure 14 shows the Settings interface with two phases: Room management where the user can add/delete rooms, cameras, contact detection module and record custom sound alert, where the user can add/delete rooms, cameras, and the contact detection module and record custom sound alerts. On the other hand, the Face enrollment is where the user can add/delete registered users.

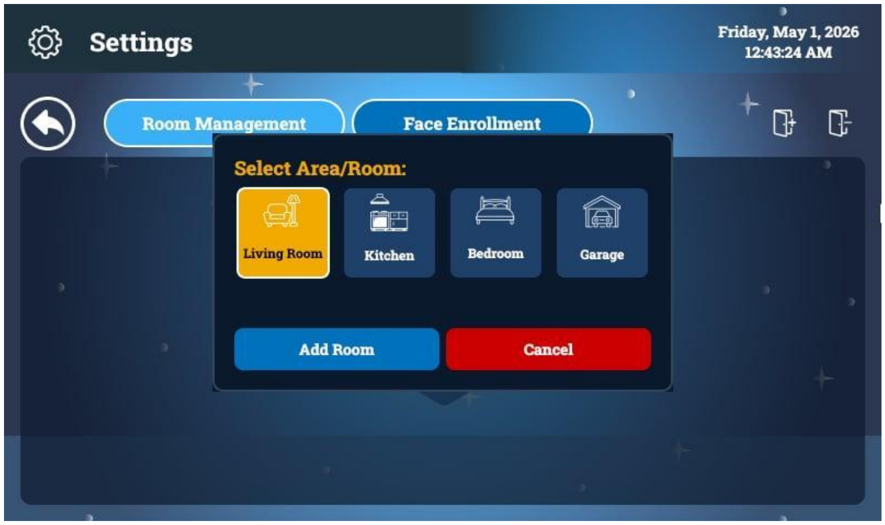
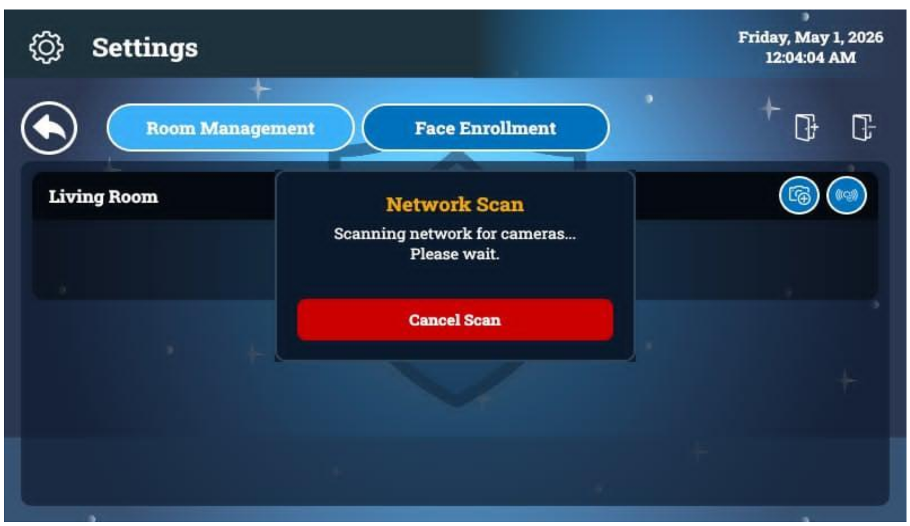


Figure 15. Hybrid Home Security System Adding Room

Figure 15 shows how the addition of a room works. By clicking the door with the plus icon, the user will be able to choose a room to be added. As for the door with the minus icon, it enables the user to delete an existing room.



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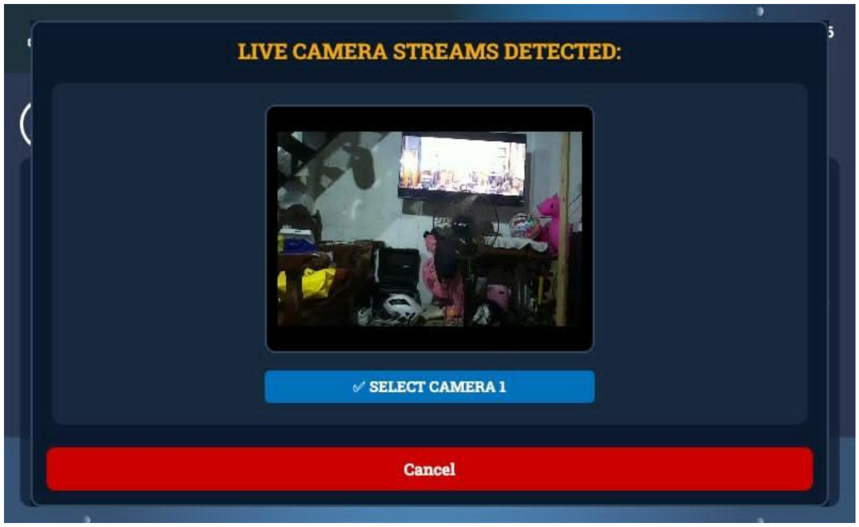
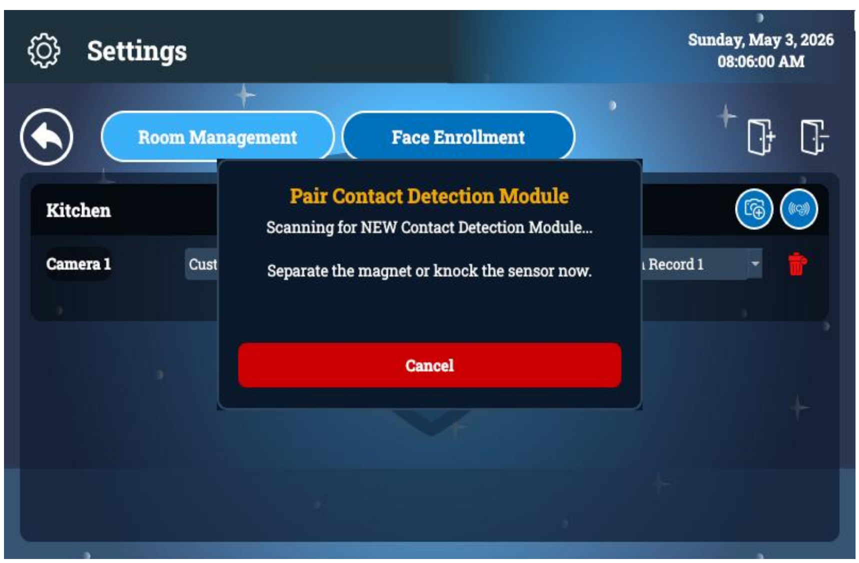


Figure 16. Hybrid Home Security System Adding Camera

Figure 16 shows how adding a camera in a room works. By clicking the camera with the plus icon, it allows the user to scan a camera nearby. If a camera is detected, the user will be able to select the camera and add the in the room.



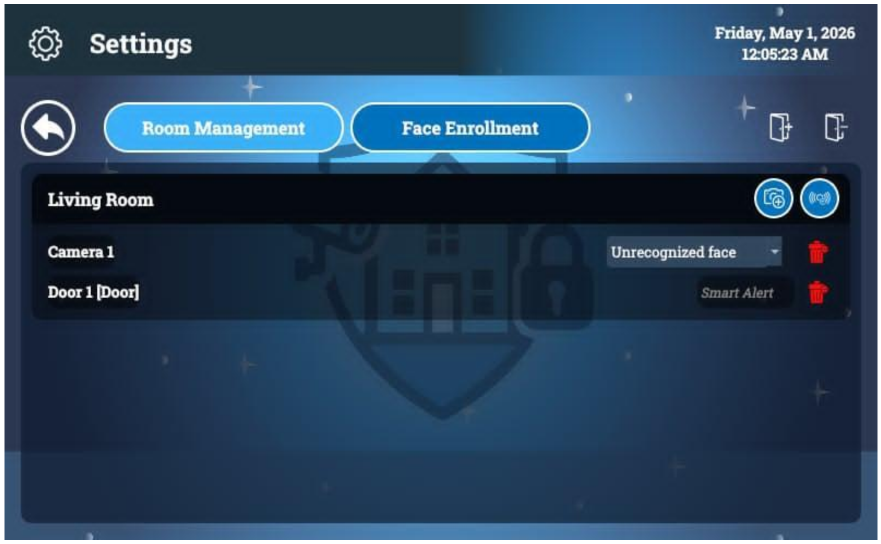


Figure 17. Hybrid Home Security System Adding Contact Detection Module

Figure 17 shows how the addition of the Contact Detection Module works. By clicking the sensor with the plus icon, it starts pairing with the Contact Detection Module that is nearby. To pair successfully, a contact interaction needs to be done. After this, the user will be able to choose where the contact detection module will be placed between the door or the window.

Figure 18. Settings Face Enrollment Interface

Figure 18 shows the second phase of the Setting, which is the Face Enrollment. This is where the addition of a user can be done to be registered. By inputting the name of the person that will be added and starting to click the “Start Enrollment Scan”, it starts scanning the front , left and right views of the person. After scanning, a confirmation will appear that the person being registered is successfully enrolled in the System.

Figure 19. Hybrid Home Security System Audio Library Interface

Figure 19 shows the Audio Library of the monitor application. This is where all the sound alerts of the system are kept, such as sound alerts for door or window interactions or when an unrecognized face is seen. Also, this is where the recorded custom sound alerts by the user can be found. Each sound has a "Play" button so the user can hear and know each of the sound alerts of the System.

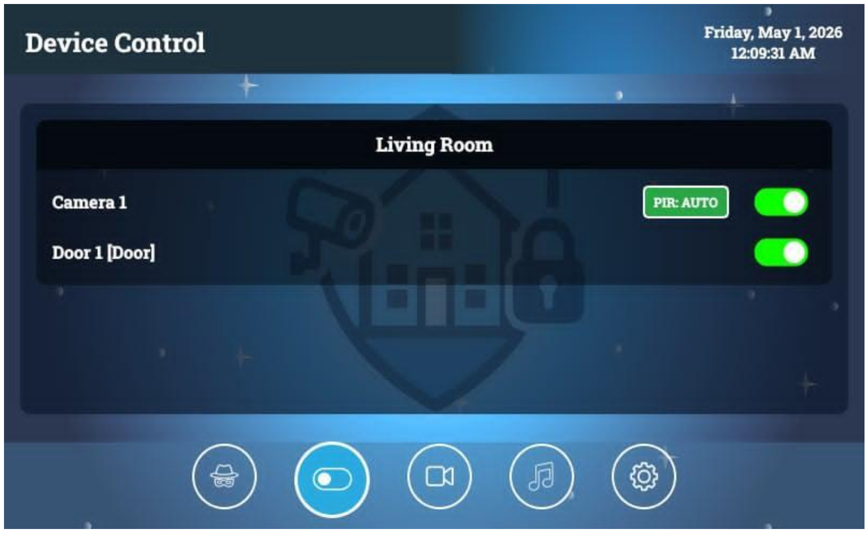


Figure 20. Hybrid Home Security System Device Control Interface

Figure 20 shows the Device Control interface to toggle each device. If the user wants to turn off or turn on again a camera, contact detection module or PIR sensor, this is where it happens, by clicking the toggle button from green color of the specific device, the button will turn to red button indicating that the device is turned off.

Figure 21. Hybrid Home Security System Live Feed Interface

Figure 21 shows the Live feed from the camera. When the camera detects a registered person, a green frame appears around the person, including their name, which means it recognizes the person detected in the camera. The battery life and state of the Contact detection Module in a door are also included in this interface so that the user will know the current state and when to charge it.

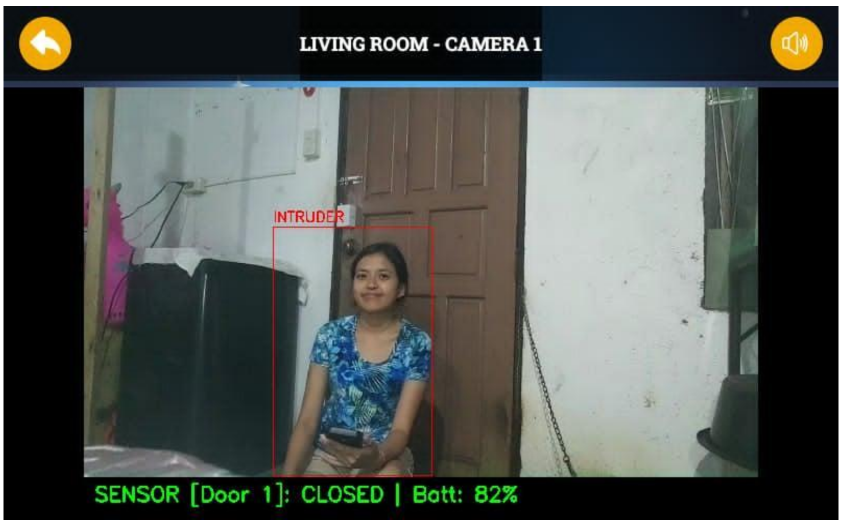


Figure 22. Hybrid Home Security System Zoom Camera for Detected Intruders

Figure 22 shows the zoomed view of the area of the camera where an intruder is detected. If an intruder is detected, a red frame is placed around that person with the "INTRUDER" indicator, which means the detected person is not registered in the System.

Figure 23. Hybrid Home Security System Captures Images of Detected Intruder

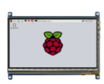
Figure 23 shows the interface display for the captured images of the detected intruder in the system. It includes the time and date when the intruder is detected so that the user will know the face of the detected intruder.

Hardware Implementation

The hardware architecture for this was assembled using components selected for their compatibility and reliability. These parts were acquired from certified electronics distributors to ensure high performance during operation.

The major hardware elements implemented into the system are detailed in the table below. These components have been specifically selected to address the needs of the Hybrid Home Security system.

Table 7. Major Components of the Hybrid Home Security

Material	Description for the System Usage	Specification
Raspberry Pi 5 	Main controller for AI, stores faces, and controls the screen.	Processor: 2.4GHz Voltage: 5 V RAM: 8 GB
Raspberry Pi Zero 2W 	Wireless node that streams video to the main Pi 5.	Processor:1.0GHz Voltage: 5V RAM: 512MB
Raspberry Pi Camera Module 3 	Attached to the Raspberry Pi Zero to watch different areas	Resolution: 11.97 MP Voltage: 3.3V Feature: Autofocus
ESP32 Super Mini 	Sends quick wireless alerts when sensors are triggered.	Processor: 32-bit RISC-V Voltage: 3.3V
Raspberry Pi 7-inch Touch Screen 	The main display for the user dashboard.	Resolution: 800 x 480 RGB Voltage: 5V Current Draw: ~500mA
Magnetic Reed Switches 	Detects when a door or window is opened or closed.	Category: Contact Sensor Logic: Normally Open (NO)
MPU6050 	Senses vibration on the door/window. .	Voltage: 3V to 5V Sensors: 3-axis Gyroscope + 3-axis Accelerometer
PIR HC-SR501 	Detects human movement to wake up the system.	Category: Motion Sensor Voltage: 5V to 20V Range: 3m to 7m
8ohm 3w Speaker 	Plays loud sound alerts and customizes alerts for the user.	Impedance: 8 Ω Power Rating: 3 Watts

Hardware components that were used in developing the System are outlined in Table 7 below. These particular components were chosen because they complement each other in executing security-related functions. The Raspberry Pi 5 acts as the main processing component, while the camera and contact sensors are responsible for identifying any intruders. Moreover, incorporating the touch screen ensures that the system is easy to control without requiring a separate computer.

The minor hardware components that were incorporated in the system include the ones listed below. The inclusion of these peripheral components is critical to extend surveillance capacity through remote nodes, issue audible alerts, and ensure ideal temperatures for the central processing unit. These components were selected based on their compactness and energy efficiency.

Table 8. Minor Components of the Hybrid Home Security

Table 8.1 Power Management Components

Material	Description for the System Usage	Specification
Lithium-ion Battery 	Stores electricity for the system to work wirelessly.	Voltage: 3.7V Full Charge: 4.2V Cut-off : 2.5V 3.0V
IP2312 	Charges the Lithium battery safely via USB.	Input: 4.5V – 5.5V / Charge: 1A
TP4056 Module 	Battery charger for the Lithium battery	Input: 4.5V - 5.5V Charge Current: 1A Full Charge: 4.2V
BMS 1S 12A 	Safety Controller of the battery	Configuration: 1S (3.7V) Overcharge: 4.25V - 4.35V
MT3608 	Voltage booster for the Facial Recognition node.	Input Voltage: 2V – 24V Output Voltage: Up to 28V
TPS61088 	High-power booster for the Pi Zero and speaker.	Input: 2.7V – 12V Output: 4.5V – 12.6V
MAX17043 	Tracks battery life percentage for the contact detection module..	Interface: 12C Function: Reports Battery %
Power Supply 	Main plug-in power for the Pi 5.	Input: 100V – 240V AC Output: 5.1V DC / 5.0A
Nut Female DC Power 	The main power port on the case where the adapter is plugged in.	Type: Female DC Jack
Power Adapter 	Provide the main power for the camera module.	Type: AC to DC Adapter Output: 5V / 3A Plug: DC Barrel

The components that deal with most of the electricity within the Hybrid Home Security System are displayed in Table 8.1. The components ensure that the Hybrid Home Security System works through wireless technology powered by batteries while ensuring that the battery stays charged, all the while supplying steady power to the Raspberry Pi 5.

Table 8.2 Audio and Visual Components

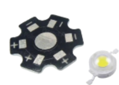
Material	Description for the System Usage	Specification
Red LED 	Visual Indicator of the system to know if it is active or not.	Voltage: 1.8V - 2.2V Current: 20mA
1w LED Bead 	Spotlight of the camera to see faces even when the surroundings are in low light.	Voltage: 3.0V - 3.4V Luminous Flux 100-110 Lumens:
PAM8403 	Audio amplifier for the system's sound alerts.	Operating Voltage: 2.5V – 5.5V) Output Power: 3W
Web Camera 	Use to register authorized user faces into the database.	Resolution: 1920 x 1080 Connection: USB
USB Microphone 	Allows users to record custom alerts.	Sensitivity: -30dB ± 3dB
USB Aux mic Adapter 	Connects the speaker to the Raspberry Pi.	Connectors: USB Type-A to 3.5mm Jack
Aux Audio Cable 	Connects the PAM8403 Amplifier to the 3W Speaker	3.5mm Jack Shielded Cable

Table 8.2 indicates the components responsible for the system's audio and video. These include lights that enable operation in low-light environments, the webcam that registers facial features, and the microphone for recording user-defined audio notifications.

Table 8.3 Circuit Protection and Connection

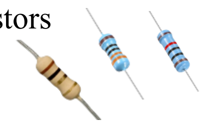
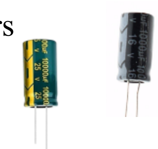
Material	Description for the System Usage	Specification
Resistors 	Limits the current to protect the LEDs and modules.	Values: 330 ohms 10k ohms, 1 ohm 0.5 watt
Capacitors 	Filters electrical noise for clean power.	Types: Low ESR Electrolytic and Radial Electrolytic Voltage: 10V - 16V Capacitance:1000uF
SS54 5A Diode 	Ensures power flows in only one direction.	Type: Surface Mount Current: 5.0 Amperes Peak Reverse Voltage: 40V
LD06AJSA 	Keeps the high-power LED bright and safe.	Category: Constant Current Input Voltage: 2.8V – 6V
Stranded Wires 	Flexible paths for power and sensor signals.	Material: Tinned Copper

Table 8.3 represents the elements responsible for the safety of the circuit and the pathways used by the electronics. Resistors and capacitors are the elements responsible for protecting the sensitive components from excess heat or noise, whereas wires and diodes are responsible for ensuring safe electrical flow through proper channels.

Table 8.4 Hardware Mounting and Storage

Material	Description for the System Usage	Specification
Micro SD Card 	Stores the OS, Python code, and face database.	Storage: 32GB Speed: Class 10
Universal PCB 	Holds components and provides circuit paths.	Category: Substrate / Interconnect Material: FR-4
Pin Header 	Used to easily connect sensors and wires.	Pitch: 2.54mm (Standard) Type: Double Row / Male
Active Cooler 	The Fan and Heatsink for the Raspberry Pi 5.	Category: Thermal Management Input Voltage: 5V Control: PWM (Pulse Width Modulation)
OTG Micro USB 	connects the USB Aux Mic Adapter to the Raspberry Pi Zero.	Type: On-The-Go (OTG) Adapter Interface: Micro-USB (Male) to USB-A (Female)
Rocker Switch 	The main power switch of the system. A button to turn on or off each module.	Type: SPST (Single Pole Single Throw) Rating: 6A 250V AC / 10A 125V AC
Battery Holder 	Holds the battery in place and connects the battery's power to the circuit	Material: ABS Plastic / PC

Components described in Table 8.4 are those that revolve around the physical design and data of the system, such as the SD card where all AI-related programs and faces are stored, the fan for cooling the Raspberry Pi 5, and the physical buttons and mounts to organize the hardware inside the case.

Table 8 below indicates the less significant yet crucial components that ensure the proper operation of the Hybrid Home Security System. Although the primary controllers manage the brain of the system, the secondary components ensure that the processes of operation take place smoothly. For instance, the power modules, including the TP4056 and BMS, provide the battery charging process, allowing the device to have a long-lasting battery life for wireless operation. On the other hand, the resistors and capacitors filter the electric current to avoid any damage to the critical components, ensuring that the audio alerts operate without generating noise. Finally, the diodes and switches allow users to control the power supply to the device while ensuring that the electric current flows in the appropriate direction.

Software Implementation

The development tools and programming environments used for this project were obtained from the official developer repository and from open-source resources. This choice of resources was made to make sure that they were compatible with the hardware structure and to serve as the foundation of the system.

The following table describes the main software packages that were used during the development of the device. These packages played a crucial role in setting up the operating environment, defining the control flow, and controlling the communication process between the security sensors and face recognition algorithms.

Table 9. Software Applications of the Hybrid Home Security

Material	Description for the System Usage	Specification
Raspberry Pi Imager	A tool to install the OS and set up network settings for the Pi boards.	Version: 1.8.5+ Compatibility: Windows 10/11
Python	The main language used to code the system logic and hardware.	Version: 3.11.2 PyQt6, RPi.GPIO
Thonny	The IDE used to write, edit and run the interface of the system.	Version: 4.1.7
OpenCV	The library used to process the camera feed and handle images.	Version 4.x
Ultralytics	Used for the detection of people in the camera view.	YOLO Model Python SDK
InsightFace	Used to locate faces and recognize if they are registered users.	Models: SCRFD (Face Detection) and ArcFace (Face Recognition;
MobileFaceNet	A lightweight CNN works with ArcFace to quickly read the unique parts of a face.	Architecture: MobileFaceNet Format: ONNX Runtime

As shown in Table 9, the following software was employed as the intelligence of the system. Raspberry Pi Imager was used to install the operating system, whereas Python and OpenCV were used as the primary language and base for handling hardware and videos, respectively. For security purposes, Ultralytics (YOLO) is deployed to detect humans, while InsightFace employs SCRFD and ArcFace to identify known individuals.

Project Structure/Organization

The following paragraphs explain the process through which the Hybrid Home Security System was designed and structured. This includes the process through which the components were integrated, as well as how the system was programmed to function for contact detection and facial recognition.

Programming

This section shows the necessary codes that are used to program and create the functionalities of the Hybrid Home Security System.

```
//Calibration Threshold for DOOR
constexpr float MIN_WIGGLE_VAL = 0.08;
constexpr float MIN_KNOCK_VAL = 0.5;
constexpr float FORCE_HIT_VAL = 5.0;
constexpr unsigned long HIT_COOLDOWN = 150;
constexpr unsigned long RESET_TIMEOUT = 3000;

//Wiggling Energy Accumulator Logic
float vibrationEnergy = 0.0;
constexpr float WIGGLE_ENERGY_THRESHOLD = 1.5;
constexpr float ENERGY_DECAY_RATE = 0.02;
unsigned long lastDecayTime = 0;

//STATES
enum SystemState { CLOSED = 0, OPENED = 1, KNOCKING = 2, WIGGLING = 3, FORCING = 4 };
const char* stateNames[] = { "CLOSED", "OPENED", "KNOCKING", "WIGGLING", "FORCING" };
```

Figure 24. Calibration Thresholds and State Definition for the Door

This code sets the sensitivity and rules for the door interactions. The first part, which is the calibration threshold defines the exact numerical values needed to distinguish between the different types of interactions such as wiggling, knocking and forcing. These calibrated values were determined by averaging the threshold data collected from testing different types of doors. The second part, the wiggling energy accumulator, contains the logic used to track the vibration over time to detect a silent intrusion. Lastly, the states section of the code defines the five possible conditions of the door which are the closed, opened, knocking, wiggling and forcing.

```
//calibration threshold for window
constexpr float MIN_WIGGLE_SQR = 0.5;
constexpr float MIN_KNOCK_SQR = 15.0;
constexpr float FORCE_HIT_SQR = 100.0;
constexpr unsigned long HIT_COOLDOWN = 150;
constexpr unsigned long RESET_TIMEOUT = 3000;

// Wiggling energy accumulator
float vibrationEnergy = 0.0;
constexpr float WIGGLE_ENERGY_THRESHOLD = 20.0;
constexpr float ENERGY_DECAY_RATE = 0.5;
unsigned long lastDecayTime = 0;
```

Figure 25. Calibration Thresholds for the Window Interactions

This code sets the sensitivity for the window interactions. Similar to the door code, it uses a calibration threshold to distinguish between different types of interactions such as wiggling, knocking and forcing. The above values were derived by calculating the average of the thresholds obtained through testing the windows. Nevertheless, the thresholds used in the code above are relatively higher compared to the door thresholds since window glass vibrates easily, and the code has a cooling period and resetting timer for accuracy purposes. Finally, the code analyzes vibration energy over time in order to determine whether there is a possible attempt at intrusion by prying open the window.

```python
def pir_worker():
    global system_enabled, pir_enabled, ai_trigger
    while True:
        pir_detects_motion = (GPIO.input(PIR_PIN) == 1)

        with state_lock:
            if system_enabled:
                if (pir_enabled and pir_detects_motion) or ai_trigger:
                    GPIO.output(LED_PIN, GPIO.HIGH)
                else:
                    GPIO.output(LED_PIN, GPIO.LOW)
            else:
                GPIO.output(LED_PIN, GPIO.LOW)
```

Figure 26. Code Snippet for PIR and AI Integration

This is the code that acts as the security node. The code ensures that the LED will be switched on only in cases where it's absolutely necessary. This is because the code considers two parameters at once: whether or not a physical motion sensor (PIR) detects any movement and whether the artificial intelligence (AI) in the main hub detects any movement too.

```python
def _check_admin_registration(self):
    db_path = AppConfig.DB_FILE
    has_admin = False

    if os.path.exists(db_path):
        try:
            with open(db_path, 'rb') as f:
                db = pickle.load(f)
                if "Admin" in db:
                    has_admin = True
        except:
            pass

    if not has_admin:
        from admin_setup import AdminSetupDialog
        dialog = AdminSetupDialog()
        dialog.exec()
```

Figure 27. Code snippet for System Initialization and Administrator Authentication

This acts as the security for the system. The security system itself needs to first verify, in a local database, whether an administrator account exists. In case the system is completely new, then the user is forced to go through the facial enrolment process. This ensures that the security of the system starts right from the beginning and cannot function without an administrator's face.

```python
def run(self):
    self.is_running = True

    if not self.known_faces:
        time.sleep(0.5)
        self.authorized.emit("Admin Bypass (No Faces Enrolled)")
        return

    cap = AsyncCamera(self.camera_index, AppConfig.CAM_WIDTH, AppConfig.CAM_HEIGHT, warmup_reads=5)
    if not cap.start():
        self.failed.emit("Camera initialization failed.")
        return

    self.ai_busy = False
    self.latest_result = None

    def run_ai(frame_to_process):
        self.ai_busy = True
        try:
            with self.ai.inference_lock:
                faces, kpss = self.ai.face_detector.detect(frame_to_process, max_num=1, metric='default')
                if faces is not None and len(faces) > 0:
                    box = faces[0].astype(int)
                    kps = kpss[0]
                    face_obj = _DummyFace(kps=kps, bbox=box)
                    self.ai.face_recognizer.get(frame_to_process, face_obj)

                    if face_obj.embedding is not None:
                        name, score = self._match(face_obj.embedding)
                        self.latest_result = (box, name, score)
                        return
                self.latest_result = None
        except Exception:
            pass
        finally:
            self.ai_busy = False
```

Figure 28. Code snippet for Biometric Access Control

This coding is intended for security purposes in the surveillance application. It employs asynchronous threading to conduct the AI facial recognition on the live video without making the interface lag. The main purpose of the code here is to check whether the face in front of the camera belongs to the admin in order to delete a user.

```
def process_frame(self, frame, run_ai=True):
    if frame is None or not AI_AVAILABLE or not self.yolo_model:
        return frame

    curr_time = time.time()
    TRACK_LOSS_TOLERANCE = 15
    VISUAL_LOSS_TOLERANCE = 1.0

        avg_brightness = frame.mean()
        if 15.0 < avg_brightness < 70.0:
            ai_frame = cv2.convertScaleAbs(frame, alpha=1.3, beta=40)
        else:
            ai_frame = frame

        self._run_yolo_async(ai_frame.copy())

        with self.ai_lock:
            ai_results = self.latest_ai_results
            self.latest_ai_results = None

        if ai_results is not None:
            unassigned_boxes = list(ai_results)
            active_ids = []

            with self.tracker_lock:
                for tid, person in list(self.tracker.items()):
                    if not unassigned_boxes:
```

Figure 29. Code snippet for Intruder Detection and Sound Alert

This is the core security logic. First, it tracks a person. Second, it waits for them to look at the camera. Next, it identifies them. If the person is "Unknown" (intruder), it automatically captures a high-quality photo evidence to the system's local storage and triggers an immediate alert. This ensures that there is a record of an unauthorized person while the system simultaneously broadcasts a sound alert.

```python
with self.controller.data_lock:
    for name, data in self.controller.sensors_config.items():
        if not data.get('active', True): continue

        config_val = str(data.get('ip', '')).strip().upper()
        config_node_id = str(data.get('pin', '')).replace("GPIO_", "").strip()

        is_mac_match = (config_val == incoming_mac and incoming_mac != "")
        is_ip_match = (config_val == incoming_ip.upper() and incoming_ip != "")

        if (is_mac_match or is_ip_match) and (config_node_id == incoming_node_id):
            match_found = name
            break
```

Figure 30. Code snippet for Contact Detection Module to Device Authentication and Mapping

This code is the interaction verifier of the contact detection module. When a sensor sends a message to the Hub, the Hub needs to know exactly which device it is. This logic checks two things also: first, it verifies the network address (MAC or IP) to ensure the message is coming from a known source, and second, it checks the unique node ID to confirm it matches the device configuration stored in the system's database. If everything matches, it maps the incoming data to the correct room and device name.

```python
if state in ["OPENED", "KNOCKING", "WIGGLING", "FORCING"]:
    if sens_sound != 'Default':
        specific_sound = sens_sound
    else:
        state_lower = state.lower()
        if state_lower == "opened": state_lower = "open"
        elif state_lower in ["forcing", "breaking"]: state_lower = "force attempt"
        specific_sound = f"{placement_str} {state_lower}.mp3"

    self.broadcast_alarm_to_nodes(triggering_cam_name=None, specific_sound=specific_sound)

if target_cam and state in ["OPENED", "FORCING", "WIGGLING", "KNOCKING"]:
    self.alert_triggered.emit(target_cam, f"SENSOR TRIGGER: {state}")
```

Figure 31. Code snippet for Automated Audio Response Logic

This coding serves as the answer engine of the contact detection unit in the whole process. Rather than having a general beep sound played from all the contact detectors that are triggered, this particular code determines its exact physical location, such as door and window, as well as the status of the intrusion, including whether it is being opened, knocked on, or forcefully broken into. This will command the remote speaker node to automatically play the relevant sound file.

```
def _match(self, feat):
    feat /= np.linalg.norm(feat)
    best_name = "Unknown"
    best_score = getattr(AppConfig, 'FACE_MATCH_THRESHOLD', 0.50)

    for name, k_feat in self.known_faces.items():
        sim = np.dot(feat, k_feat)
        if sim > best_score:
            best_score = sim
            best_name = name

    return best_name, best_score
```

Figure 32. Code snippet for Biometric Identity and Matching Logic

The above is the engine for recognition. The system when presented with a face, compares the face against all faces in its database by a process of computing similarity levels. Where the level of similarity passes the preset threshold, then it outputs the user name that matches it. Otherwise, where no face in the database passes the threshold of similarity, it will output Intruder and an audible alarm.

Assembly

This section presents the actual integration of the system's hardware components into their respective enclosures. It discusses the placement and connection of the Contact Detection Module, Camera Module, and Raspberry Pi 5 with a 7-inch touchscreen, showing how each component was fixed, wired, and secured to ensure proper operation of the Hybrid Home Security System

A. Assembly of Contact Detection Module

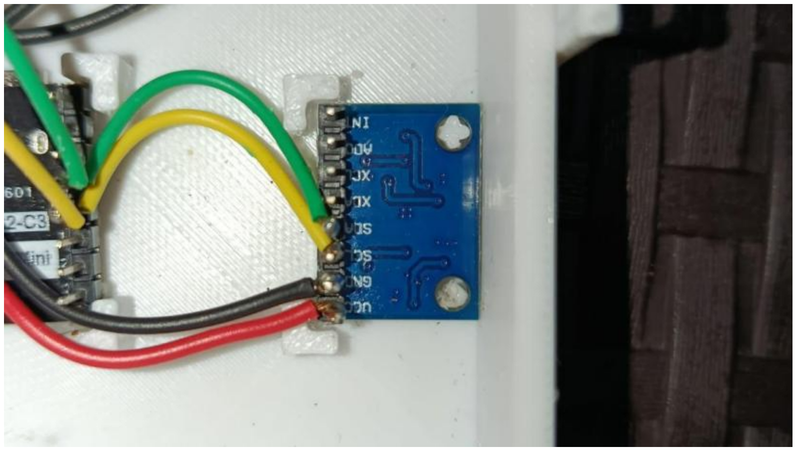


Figure 33. Assembly of MPU6050 Motion Sensor

Figure 33 shows how the MPU6050 is positioned inside the Contact Detection Module. The sensor is placed in a fixed position with the built-in pillars and screw hole to support it inside the case. These pillars or dividers act as the steady base to hold the sensor tightly to prevent it from moving or falling out in the case. Also, it shows here that its VCC, GND, SCL and SDA are connected directly to the ESP32 Super Mini.

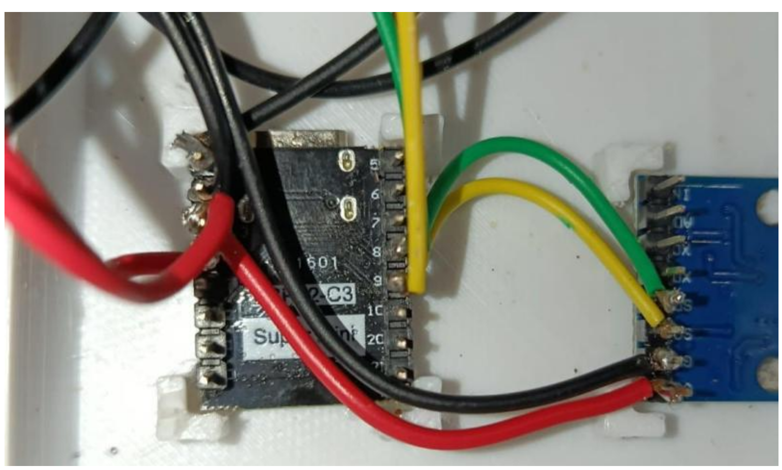


Figure 34. Assembly of ESP 32 c3 Super Mini

Figure 34 shows the ESP32 Super Mini positioning inside the case. Just like the MPU6050, the ESP32 Super Mini is also held firmly in place by the pillars or dividers to ensure that the microcontroller stays in a fixed position. The ESP32 Super Mini is wired to several parts, such as the MPU6050 for vibration sensing, the Magnetic Reed Switch for door/window state and a battery fuel gauge to monitor the battery life of the module and also a switch for the power. Since it is in a fixed position, this can send wireless alerts to Raspberry Pi 5 without problems.

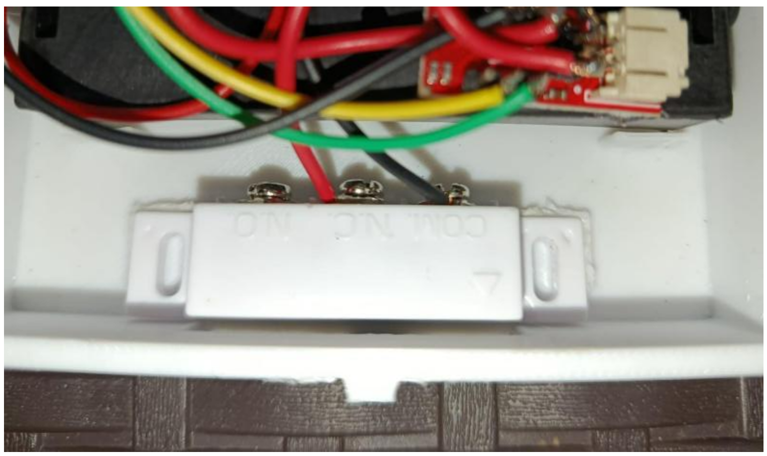


Figure 35. Assembly of Magnetic Reed Switches

Figure 35 shows the position of the Magnetic Reed inside the case, just like the MPU6050 and ESP32 Super Mini, the Magnetic Reed Switch also has pillars of its own to fix its position. This is wired directly to the ESP32 Super Mini which allows the System to immediately know if the door is being opened.

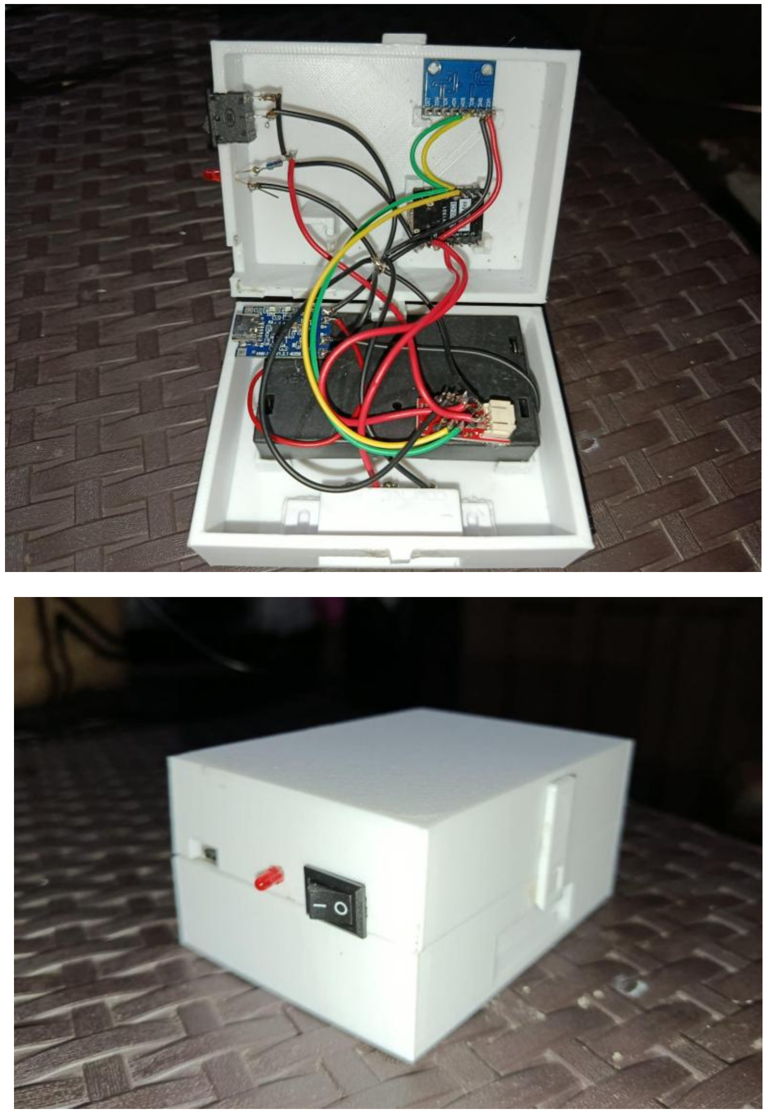


Figure 36. Assembly of Contact Detection Module

Figure 36 shows the final assembly of the Contact Detection Module, both when opened and closed. Inside the case, all components, including the battery system, MPU6050, Magnetic Reed Switch and ESP32 Super Mini, are held in place by having their own pillars to prevent them from falling out of their positions. On the other hand, outside of the case, the rocker switch and red LED indicator are very accessible, allowing the user to turn on and off the module and see its state status, besides. Besides the red LED, the user also has access to the Type-C charging module for battery charging. This compact white case is designed to be easily mounted on the door or window, which keeps the internal wires protected and gives the system a clean look.

B. Assembly of Camera Module

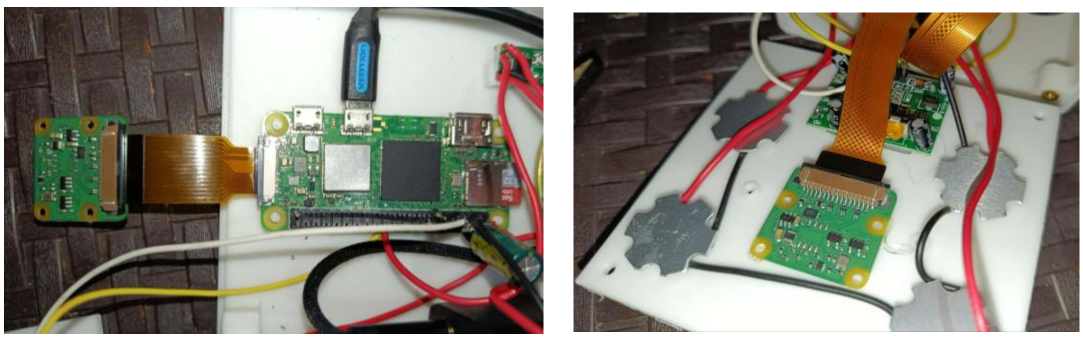


Figure 37. Assembly of Raspberry Pi Zero 2W with Camera Module 3

Figure 37 shows the assembly of the Raspberry PI Zero 2 W with Camera Module 3. Both of them are securely placed on their designated pillars or dividers, the Raspberry Pi Zero 2W on the base of the case and the Camera Module 3 in the front of the case to keep them stable. The Camera Module 3 is connected to the Raspberry Pi Zero 2W using a flat ribbon cable, alongside the other system connections.

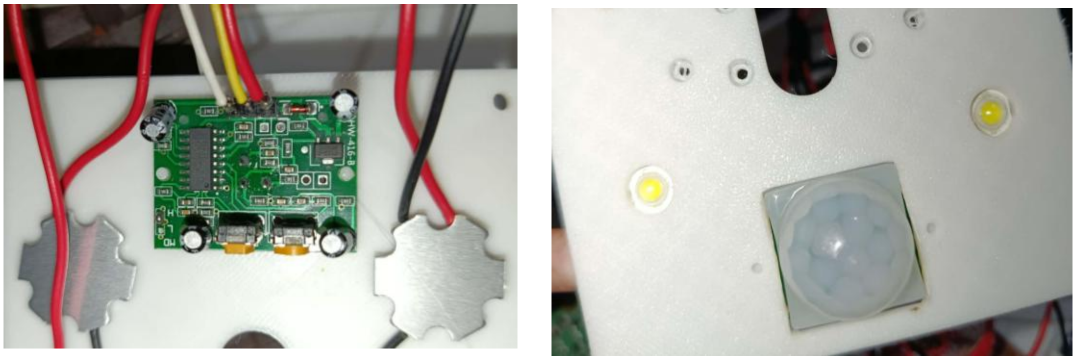


Figure 38. Assembly of PIR HC-SR501

Figure 38 is the assembly of the PIR HC-SR501 together with the LED beads. The left side image shows the back side of the PIR HC-SR501, which displays the sensor's circuit board and wire connections. On the other hand, the right side image shows the front side of the sensor in the front view of the case. Both of these are also designated to their own position in the case so that they are in place.

Figure 39. Assembly of the Speaker

Figure 39 shows the assembly of the 8-ohm 3-w speaker inside the base of the case. The red and black wires attached to the back of the speaker are connected directly to the PAM8403 audio amplifier to make the audio louder. Since this is included in the Camera

Module, it is certain that the alerts that will be played when an unrecognizable contact or intruder is detected will be hearable in each room that has a camera.

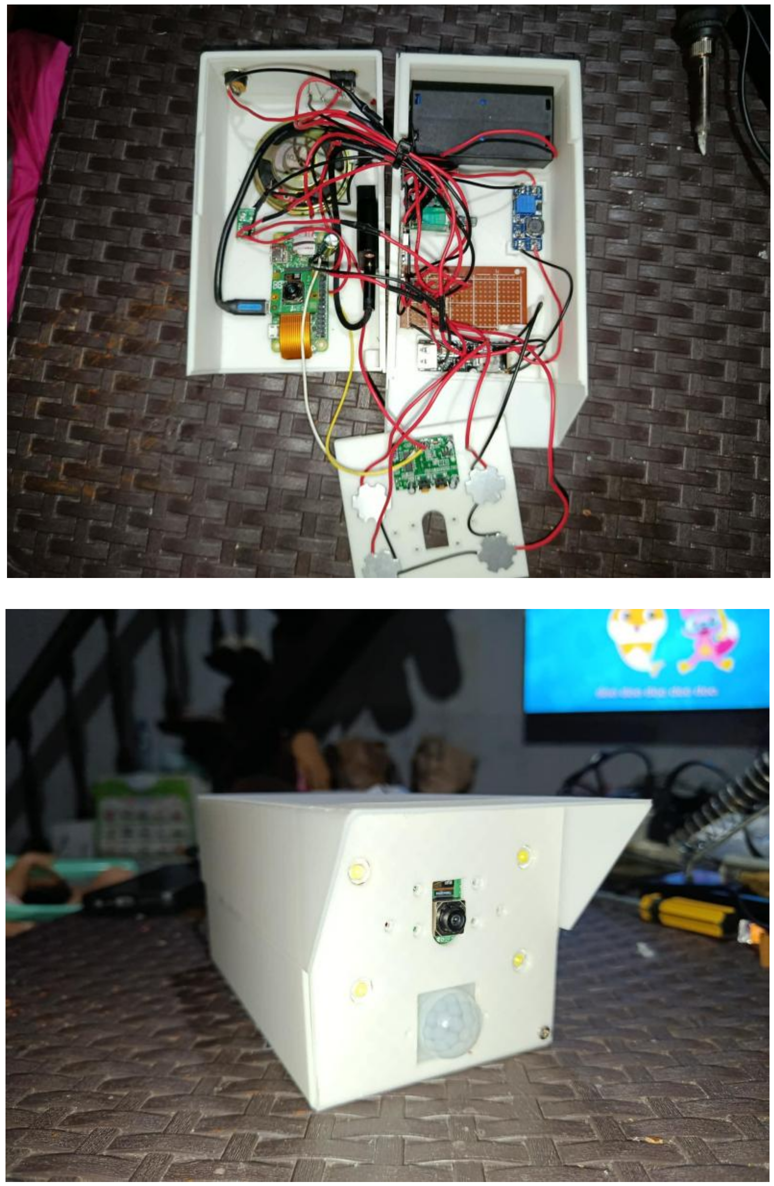


Figure 40. Assembly of the Whole Camera Module

Figure 40 presents the complete assembly of the whole Camera Module. The top image shows the look of the component connections inside the case. The bottom image shows the look of the Camera Module from the outside, it shows the camera lens, PIR motion sensor and the LED beads that are used for low light surroundings.

C. Assembly of Raspberry Pi 5 and 7-inch Touchscreen

Figure 41. Assembly of Raspberry Pi 5 with 7-inch Touchscreen

Figure 41 illustrates the physical integration of the Raspberry Pi 5, with its primary visual interface, the 7-inch DSI touchscreen. As shown in the top image, the Raspberry Pi 5 is mounted directly onto the rear side of the display touchscreen. This compact design is achieved by utilizing a flat flexible ribbon cable, which handles both data transmission and power. On the other hand, the bottom image shows the touchscreen securely housed within its case. By combining the processing power of the Raspberry Pi 5 with a touch-friendly interface, the system allows the user to easily monitor live camera feeds, manage security settings, and interact with the system's dashboard in easier way.

Evaluation

This section presents the testing and evaluation of the Hybrid Home Security System. The primary focus of this evaluation is to determine the system's effectiveness and efficiency as a solution for the intrusions that often occur without the resident's awareness

Table 10 - 16 presents the effectiveness testing of the Contact Detection Module when installed on different types of doors and windows. The module was tested using four common interactions: opening, knocking, wiggling, and forcing. Each interaction was conducted in 10 trials to determine how effectively the system could detect and classify door and window activities. The results served as the basis for evaluating the effectiveness of the contact module under different materials.

Table 10. Contact Detection Module Effectiveness in Standard Wooden Panel Door

Standard Wooden Panel Door Interaction	**Trials**										**Percentage**
	1	2	3	4	5	6	7	8	9	10	
Opened	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Knocking	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Wiggling	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Forcing	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%

Table 10 shows the effectiveness of the contact detection module on a standard wooden pane door. The researchers tested the hardware 10 times for four different interactions, which are opening, knocking, wiggling and forcing the door. Out of 40 trials, the contact detection module got a 100% success rate in testing. For the "opened" interaction in all 10 trials, the system properly detected the door when it was being opened. As for the other interactions, which are the knocking, wiggling, and forcing, each trial of these interactions also gives a 100% success rate, as the contact detection module stayed in "closed" state if an interaction is not triggered in the contact detection module.

Table 11. Contact Detection Module Effectiveness in Mahogany Panel Door

Mahogany Panel Door Interaction	Trials										Percentage
	1	2	3	4	5	6	7	8	9	10	
Opened	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Knocking	✓	✓	✓	x	x	✓	x	✓	✓	✓	70%
Wiggling	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Forcing	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%

Table 11 shows how the contact detection module works on a mahogany panel door. The trial for “opened”, “wiggling”, and forcing achieved the 100% success rate when testing it. While for the “knocking” interaction, it only achieved 70% of success rate as the other 30% shows the performance variability during the testing because there’s an instance where the knocking interaction is detected as wiggling only, as the materials used for the door are mahogany.

Table 12. Contact Detection Module Effectiveness in Molded Panel Door

Molded Panel Door Interaction	Trials										Percentage
	1	2	3	4	5	6	7	8	9	10	
Opened	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Knocking	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Wiggling	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Forcing	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%

Table 12 shows the representation of how the trials for the contact detection module in the molded panel door also achieved 100% success rate out of 40 trials. For the “opened” interaction , in all 10 trials, the system also successfully detected the interaction when the door is being opened, since as the pair of magnetic reeds is far away from each other, it will sense that the door is opened. For the other interactions, it also got 100% successful rate, but because it was tested in a molded panel door, which is lightweight and hollow, it shakes and vibrates easily, but despite that, the 100% success rate proves that the contact detection module also stayed in “closed” state until an interaction is made.

Table 13. Contact Detection Module Effectiveness in Sliding Window

Sliding Window Interaction	**Trials**										**Percentage**
	1	2	3	4	5	6	7	8	9	10	
Opened	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Knocking	✓	✓	x	✓	✓	✓	✓	✓	✓	✓	90%
Wiggling	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Forcing	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%

Table 13 represents how the contact detection module works in a sliding window. The hardware was tested 10 times each for the four interactions. The system performed well by achieving an almost 100% success rate except for one 1 trial during the knocking test. For “opened”, “wiggling” and “forcing” interaction, the contact detection module got 100% success rate while testing, during the “knocking test”, out of 10 trials, only 9 were successful, as it only has 90% success rate, and the other 1 test or the 10% is misclassified as a wiggling interaction. This result likely happened because when knocking in a sliding window, it varies depending on how strong it is and the materials used for it.

Table 14 . Contact Detection Module Effectiveness in Jalousie Windows

Jalousie Window Interaction	**Trials**										**Percentage**
	1	2	3	4	5	6	7	8	9	10	
Opened	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Knocking	✓	x	✓	x	x	x	x	x	x	x	20%
Wiggling	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Forcing	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%

In Table 14, this is how the effectiveness of the contact detection module works in a jalousie window. Just like others, the “opened”, “wiggling” and “forcing” interaction also got 100% success rate while the knocking interaction achieved only 20% success rate, since when knocking, the system sometimes only detected it as wiggling of the jalousie window because the jalousie glass slat itself is wobbling already because of its frame.

Table 15. Contact Detection Module Effectiveness in Steel Casement Windows

Jalousie Window Interaction	**Trials**										**Percentage**
	1	2	3	4	5	6	7	8	9	10	
Opened	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Knocking	✓	✓	x	x	✓	x	✓	✓	✓	✓	70%
Wiggling	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Forcing	✓	✓	✓	✓	x	✓	✓	✓	✓	✓	90%

Table 15 shows how effective the contact detection module is in a steel casement window. For the trials of “opened” and “wiggling” interaction, just like others, a 100% success rate is achieved by the system. On the other hand, the “knocking” interaction only achieved 70% success rate, as 3 out of 10 trials were misclassified by the system as 'wiggling. Similarly, the “forcing” interaction achieved a 90% success rate, with 1 trial misclassified as 'knocking. These results show a classification overlap where the vibration patterns of an interaction share similar frequency.

Table 16. Contact Detection Module Effectiveness in Awning Window

Awning Window Interaction	Trials										Percentage
	1	2	3	4	5	6	7	8	9	10	
Opened	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Knocking	✓	x	✓	✓	✓	x	x	✓	x	✓	60%
Wiggling	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
Forcing	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%

In table 16, the "knocking" interaction only got the 60% success rate compared to the other 3 interaction that achieve a 100% success rate. The "knocking" interaction is detected as a "wiggling" interaction only by the system, as the window glass used a thick material.

Tables 17 and 18 present the effective distance testing of the Facial Recognition System under daylight and low-light conditions. The test evaluated how far the Camera Module could accurately recognize registered users and classify non-registered users as intruders. The results help determine the system's reliable operating range in different lighting environments.

Table 17. Effective Distance for Facial Recognition of Registered Users

Condition	Distance from the Camera Module (m)									
	0.5	**1.0**	**1.6**	**2.2**	**2.8**	**3.1**	**3.9**	**4.25**	**5.1**	**5.3**
Day Light	✓	✓	✓	✓	✓	✓	✓	✓	✓	x
Low Light	✓	✓	✓	✓	✓	✓	✓	✓	x	x

Table 17 shows the result of the 10 trials for the facial recognition of the camera for registered users in the system based on the distance of the person from the camera. The testing distance range is from 0.5 m to 5.3 m. The researchers tested the camera in two lighting conditions. For daylight out of 10 trials, 9 are successful with a range from 0.5 m to 5.1 m. When the registered user is 5.3 m far away, the system's facial recognition is not that effective on the registered user. In a low-light environment, the system's facial recognition is effective in recognizing registered users from a distance range of 0.5 m to 4.25 m. When the distance reaches 5.1 m, the registered user is not recognized anymore by the system, as it outputs that the person detected is an intruder. In a daylight environment, the maximum distance where the camera successfully recognized the registered user was 5.1 m; however, in a low light environment, the facial recognition of the camera could not see quite as far as the maximum successful distance dropped to 4.25 m only. Having good lighting allows the system to reliably recognize faces from further away.

Table 18. Effective Distance for Facial Recognition of Non-Registered Users

Condition	Distance from the Camera Module (m)										
	0.5	**1.0**	**1.6**	**2.2**	**2.8**	**3.1**	**3.9**	**4.25**	**5.1**	**5.3**	**5.5**
Day Light	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	x
Low Light	✓	✓	✓	✓	✓	✓	✓	✓	✓	x	x

Table 18 presents the empirical evaluation of the facial recognition pipeline when processing non-registered users across a distance range of 0.5 m to 5.5 m. The evaluation was conducted under two environmental conditions: daylight and low light. When the system detects a face that does not possess a registered feature vector in the database, it automatically classifies the individual as an unauthorized entity or intruder. From the collected data, the results show that the system is capable of detecting the intruders within an optimal distance of 5.3 m during optimum lighting conditions. In cases where low light occurs, the maximum distance that the system can effectively operate at decreases to 5.1 m, considering the anticipated loss in sensor resolution. From the trials conducted, the mean average value of the operational distance of the intruder recognition was obtained at 2.975 m.

The verification results of the Facial Recognition System for registered users can be found in Table 19 below. This table depicts how the facial recognition system verifies the identity of authorized persons using the saved facial profiles in the database without triggering any alarms in case of a successful registration of the user. In this regard, a registered user is considered to be verified if the bounding box color changes to green and the registered user's name appears.

Table 19. Registered User Identification Verification

Registered User	Verification Result	System Response
Trina	VERIFIED: Trina	No Alarm Triggered
Moimoimoi	VERIFIED: Moimoimoi	No Alarm Triggered
Biyya	VERIFIED: Biyya	No Alarm Triggered
Killian	VERIFIED: Killian	No Alarm Triggered
Trisha	VERIFIED: Trisha	No Alarm Triggered
Jericho	VERIFIED: Jericho	No Alarm Triggered
Wilma	VERIFIED: Wilma	No Alarm Triggered
Diego	VERIFIED: Diego	No Alarm Triggered
Lindsay	VERIFIED: Lindsay	No Alarm Triggered
Kimberly	VERIFIED: Kimberly	No Alarm Triggered

This can be seen from Table 19, which illustrates the behavior of the system when the AI identifies the person who is already registered in the database. When the test was done, the camera took a picture of the person and compared their face against the faces of the residents saved in the database. If the faces matched, the system would recognize them based on their name and enter into the "No Alarm" state.

Response time from sensor detection to alarm generation for the Hybrid Home Security System is presented in Table 20. It covers the response time of the Magnetic Reed Switch if there is a door opening event, the MPU6050 if there are wiggling, forcing, and knocking events, and Facial Recognition Processing for registered and unregistered users in 10 trials. The average value was calculated to determine how fast the system detects an event and generates an alarm.

Table 20. Time response from Sensor to Alarm

Sensor	**Trials (seconds)**										**Ave. Time (s)**
	1	2	3	4	5	6	7	8	9	10	
Magnetic Reed Switch	0.70	0.71	0.70	0.63	0.69	0.65	0.68	0.63	0.64	0.68	0.67
MPU6050	0.90	0.75	0.82	0.92	0.86	0.93	0.80	0.95	0.85	0.83	0.86
Facial Recognition Processing (Registered)	1.66	1.45	2.00	1.96	1.61	1.34	1.69	1.10	2.34	1.23	1.63
Facial Recognition Processing (Unregistered)	2.27	2.15	2.25	2.00	2.12	1.95	2.07	1.93	2.20	1.98	2.09

Table 20 demonstrates the time response performance of each sensor and processing module included in the Hybrid Home Security System. It illustrates the response times of each process from ten trials in seconds, together with their respective average response times. Among all modules, the Magnetic Reed Switch sensor gave the quickest response, which was on average 0.67 seconds. This implies that there will be efficient detection and an alarm when a door or a window opens. On its part, the MPU6050 sensor produced an average response time of 0.86 seconds, showing that efficient movement, vibration, and interaction were effectively detected by the system. In regard to the Camera Module, Facial Recognition for registered users took an average of 1.63 seconds in terms of average response time, while for unregistered users, the system registered an average of

2.09 seconds. There was a higher average response for unregistered individuals, since this involved some extra steps in verifying their identity.

Table 21 shows the performance of the Hybrid Home Security System under different user roles and contact interactions. The researchers conducted 10 trials involving different individuals classified as either registered users or intruders. The evaluation was performed to determine whether the system could correctly recognize authorized and unauthorized users, detect door or window interactions, and activate the sound alert only when necessary.

Table 21. System Performance Under Different User Roles and Contact Interactions

Trial No.	User and Role	Expected Camera Recognition	Actual Camera Recognition	Contact Interaction (Door/Window)	Expected Sound Alert Status	Actual Sound Alert Status	Remarks
1	Bea (Intruder)	Unauthorized	Unauthorized	Wiggling	Activated	Activated	Successful
2	Trina (Registered user)	Authorized	Authorized	Opened	Inactive	Inactive	Successful
3	Moises (Registered user)	Authorized	Authorized	Wiggling	Inactive	Inactive	Successful
4	Sheree Nicole (Intruder)	Unauthorized	Unauthorized	Knocking	Activated	Activated	Successful
5	JM (Intruder)	Unauthorized	Unauthorized	Opened	Activated	Activated	Successful
6	Trina (Registered user)	Authorized	Authorized	Wiggling	Inactive	Inactive	Successful
7	Bea (Intruder)	Unauthorized	Unauthorized	None	Inactive	Inactive	Successful
8	Bea (Intruder)	Unauthorized	Unauthorized	Forcing	Activated	Activated	Successful
9	Angela (Intruder)	Unauthorized	Unauthorized	Opened	Activated	Activated	Successful
10	Moises (Registered user)	Authorized	Authorized	Opened	Inactive	Inactive	Successful

Table 21 is the performance summary of the Hybrid Home Security System during the pre-entry trials. The effectiveness of the system is clearly revealed from the Camera Module for Facial Recognition since it could detect and recognize individuals as either "authorized" or "unauthorized". As seen in the table, the actual result of the system was 100% successful in 10 consecutive tries and triggered a sound alarm for any of the unauthorized individuals while detecting all authorized ones. The effectiveness of the Hybrid Home Security System is demonstrated through the response of the Contact Detection Module Interaction and the Sound Alert Module. For instance, during the interaction of the door with registered persons such as Trina and Moises, the alarm does not ring even when there's an interaction with the door, as the camera detects them as "Authorized". On the other hand, any contact interaction by the unauthorized subjects such as Bea, Sheree Nicole, JM, and Angela automatically triggers the corresponding sound alert, while if an unauthorized individual is detected and does not make any contact interaction, the sound alert is inactive.

Table 22 presents the evaluation of the stability test conducted over a five-hour period of continuous system operation with the camera module and the contact detection module. This test was done to measure the system speed, video quality change over time and if the functionality is correct. The table shows the response time in seconds for both video stream latency and sensor response time, and whether it is working properly.

Table 22. System Stability and Response Time over Continuous Operation

Time Interval	Video Stream Latency (sec)	Sensor Response Time (sec)	Observation
0-1 Hour	0.3	0.765	Real Time Detects individuals correctly Correctly classified contact interaction
2-3 Hours	0.5	0.92	Real Time Detects individuals correctly Correctly classified contact interaction
4-5 Hours	1.0	1.05	Slight Frame Drops Misclassification of interaction Misidentification of detected individuals

Table 22 shows that the system is highly efficient during the first 3 hours of continuous use. In the testing, between 0 and 3 hours remains very fast, ranging only between 0.8 seconds and 1.25 seconds. During that period, the system status is capable and does not misidentify the individuals or misclassify the interactions. However, the performance of the system dropped in the 4 hours. During the 4 hours, the response time of the video stream increases to 4.2 seconds and the sensor response time increases to 1.3 seconds, making the video stream and sensor response delayed, and the identification of the detected individuals in the facial recognition is misread.

These results show that while the system is highly efficient in short-term use, continuous operation for four hours causes it to lag. This slowdown is caused by processor heat from running the facial recognition algorithm, like ArcFace, continuously. To maintain its fast speed, the system needs a regular automatic restart.

Limitations and Capabilities

The following section details the capabilities representing the system's successful features and technical advantages, and the limitations identifying the environmental and technical boundaries encountered during the testing of the system.

Capabilities

Based on the evaluation results, the Hybrid Home Security System is capable of detecting common door and window interactions such as opening, wiggling, forcing, and knocking using the contact detection module. The system performs efficiently in terms of detecting any opening of doors or windows, wiggling and forced entry with almost all material types having a 100% rate of accuracy. Additionally, the system is capable of detecting registered individuals and preventing unnecessary alarms from being triggered, while unregistered users are considered to be intruders. In regard to response time, the system was able to detect and trigger an alarm in an efficient manner, with the Magnetic Reed Switch triggering an alarm at an average of 0.67 seconds, MPU6050 with 0.86 seconds, registered face detection with 1.63 seconds and unregistered face detection at 2.09 seconds. Lastly, the Facial Recognition System was found to work effectively for a distance range of 1 to 5 meters in both natural and low light, with detection being highest with one to three people detected.

Limitations

In spite of the high level of effectiveness of the system, there are some problems with the classification of certain vibration interactions including knocking. According to the results obtained, knocking interaction had been sometimes misclassified as wiggling in the case of jalousie windows, awning windows, mahogany doors, and steel casement windows. It means that the vibrations of some materials can be similar and hard for the MPU6050 sensor to differentiate. There are also some distance and lighting restrictions associated with the work of the Facial Recognition System. Though its functioning within 1 to 5 meters proved to be reliable, the system lost its efficiency beyond the said range, especially if there was poor lighting. Moreover, it works less efficiently when there are several people on the screen since the detection rate was much lower if there were four or five people in the frame. Hence, it would be better to use the system in those entry points where a limited number of people could appear in the frame of a camera.

Chapter V

SUMMARY, CONCLUSION AND RECOMMENDATION

This chapter discusses the findings of the research conducted. The findings are those obtained after analyzing, observing, and interpreting the data collected. Recommendations are then made to improve the research.

Summary

The study developed a Hybrid Home Security System with a combination of contact sensors and computer vision to make homes safer. The researchers used a Raspberry Pi 5 as the brain of the system and a Raspberry Pi Zero as a sensor node. The system was designed to identify individuals using facial recognition algorithms and to detect unauthorized contact through doors and windows using magnetic reed sensors and an MPU6050 sensor.

The data transmission in this case was carried out by sending data through the personal hotspot created by the Raspberry Pi 5 using both UDP and MQTT protocols. An alert sound was emitted on detection of a person whose face or contact was not recognized.

Evaluation of the system was done using its effectiveness and efficiency. From the findings, it was noted that the system could effectively tell apart registered residents from intruders, and the effectiveness of the contact detection module during interaction with doors and windows. From efficiency analysis, it was noted that the system was very efficient and responded to actions at very low latency times, beating the set average response time. Overall, this study came up with a practical security solution.

Findings

A successful contact detection module has been created through the use of magnetic reed sensors and MPU6050 sensors linked to the Raspberry Pi Zero. The results reveal that the detection of both door and window interactions has been achieved effectively by the sensors. Therefore, any movement at the doors and windows is successfully captured as immediate alerts are relayed to the brain from the hardware.

A successful facial recognition feature has been developed through the use of computer vision algorithms on the Raspberry Pi 5. In the initial plan, three primary processes were identified: human detection, face detection, and face recognition. YOLO was used for human detection while SCRFD was used for face detection. The recognized face was then matched using ArcFace. Through this setup, the system is able to differentiate a registered user from an intruder with the help of the Camera Module 3. Thus, the software is able to recognize facial features, which helps determine the course of action: silent or sound alerts.

From the results, it can be observed that the sound alert system works according to its design, using the provided sound alert of the system and customizing sound alerts. In the event that there is an intruder within the premises and the contact detection module is activated, the system is designed to produce certain sound alerts, which clearly indicate whether an intruder exists, thus alerting the homeowners.

It can be concluded from the experiment that the Hybrid Home Security System proved to be effective and efficient in performing all of its functions. With regard to effectiveness, the system detected door and window interactions, verified registered users, detected non-registered users as intruders, and triggered an alarm when required. Thanks to the interaction between the contact detection module, camera module, facial recognition system, sound alert system, and the control and monitoring hub, the system was able to carry out physical detection of intrusion attempts and verify users' identity. Regarding efficiency, the system demonstrated quick responses to events. The Magnetic Reed Switch showed the fastest response rate at an average of 0.67 seconds, while the MPU6050 had a response rate of 0.86 seconds on average. The processing of faces belonging to registered users took 1.63 seconds, while the processing of unregistered users' faces took 2.09 seconds. Thus, one may conclude that this home security system is able to detect events and trigger responses with minimum delays by providing active detection, facial verification, and immediate sound alert capability.

Conclusions

The research work succeeds in meeting the objectives of the research project through the design and implementation of the Hybrid Home Security System. This Hybrid Home Security System is able to design and implement a "main brain" using the Raspberry Pi 5 and a sensor node using Raspberry Pi Zero to form a Hybrid Home Security System. From the various tests conducted, the Hybrid Home Security System proves to be efficient in detecting physical contact through magnetic reed sensors and MPU6050 sensors. It is able to detect the registered users and differentiate them from the intruders through a facial recognition algorithm. It is also able to give instant feedback by producing sound alarms when an unauthorized contact and face is detected.

Recommendations

Based on the findings and conclusions presented, the following recommendations are suggested:

1. **Enhance AI Recognition Accuracy**

   Future researchers can utilize more advanced AI models and image processing techniques to help the camera see clearly in poor lighting. By training additional AI with a wider variety of images taken in different lighting conditions, future studies can ensure that the system stays consistent whether in daylight or at night.

2. **Integration of Cloud-Based Storage**

   Using cloud storage provides a much larger place to save captured images of detected intruders, so there is no need to manually delete old files to make room for others. Also, this would allow the user to easily check their security captured images and monitor their home from anywhere using the internet.

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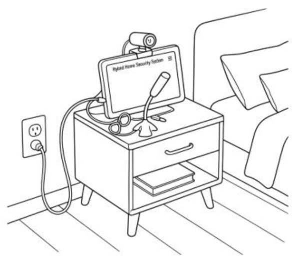
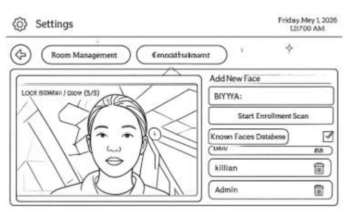
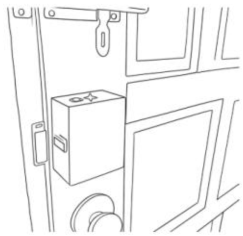
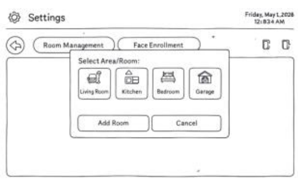
APPENDICES

APPENDIX A. User Manual

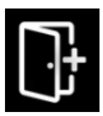
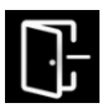
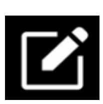
Setting up the Hybrid Home Security System

Steps	How?
1 	Position the main hub together with the web cam and microphone in the bedroom preferably beside the electrical outlet and turn on the monitor.
2 	Register the face of the Admin.
3 	Enroll face at the Settings face enrollment.
4 	Position the Camera Module mounting on the wall at the top of the outside door, preferably beside an electrical outlet and turn it on.

5 	Position the Contact detection Module in door/window and turn it on.
6 	Go back to the main hub and try adding room in Settings room management.
7 	For adding camera, click the “camera with plus” icon to detect if a camera module is near and active.
8 	For adding contact detection module, click the “sensor with plus” icon then try to make an interaction to the contact detection module in order for the system to detect if a contact module is active.
9 	The System set up is now completed.

Button Guide of Hybrid Home Security System Monitoring Application

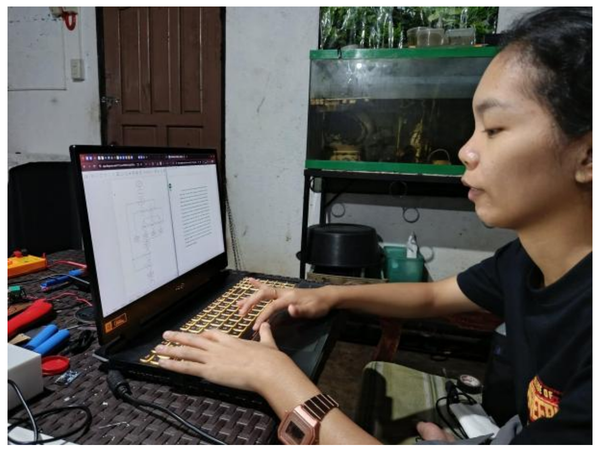
Application Buttons	Description
1 	Button for admin face registration.
2 	Button for the display interface for the captured images of detected intruder.
3 	Button for device control interface to toggle the device in a room.
4 	Button for live feed.
5 	Button for audio library to play a sound alerts.
6 	Settings button for the management of the system.
7 	Button indicator of a device when turn on.
8 	Button indicator of a device when turn off.
9 	Button for turning on/off of PIR sensor.

10 	Button for muting the sound alert.
11 	Play button for playing the sound alerts.
12 	Button in to manage the roo set up.
13 	Button for enrolling face of authorize person.
14 	Button for adding room.
15 	Button for deleting and existing room.
16 	Button for adding camera in a room.
17 	Button for adding contact detection module in a room.
18 	Button to register a face of a person.
19 	Button to delete a registered face in the database.

APPENDIX B. Documentation

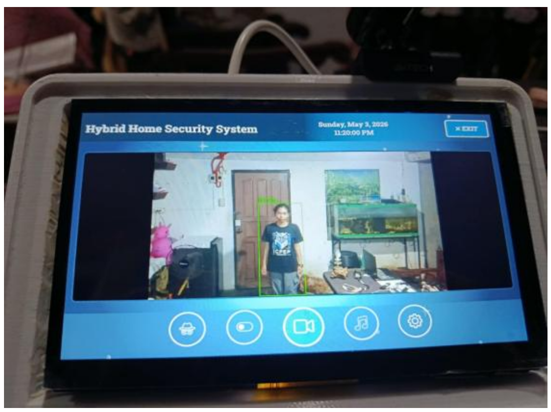
Hybrid Home Security System Designing, Coding, Assembling and Testing and Paper Making




Teranex


TUBIG


Hybrid Home Security System
Sunday, May 3, 2026
11:20:00 PM
× EXIT

APPENDIX C. Threshold Calibration Tables

Threshold Calibration for Different Types of Doors

Interaction Type	Standard Wooden Door (g)	Mahogany Door (g)	Molded Door (g)	Average Threshold (g)
Knocking	0.46	0.58	0.49	0.51
Wiggling	0.07	0.10	0.08	0.08
Forcing	4.8	5.2	4.9	5.0

This table presents the computed threshold values for door intrusion detection used in the Hybrid Home Security System. The researchers conducted 10 experimental trials on three different door types namely standard wooden door, mahogany door, and molded door. Various interaction scenarios including knocking, wiggle attempts, and forced opening were performed to obtain MPU6050 sensor readings. The average values gathered from the calibration tests served as the basis for the threshold values embedded in the system program.

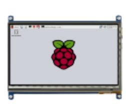
Threshold Calibration for Different Types of Windows

Interaction Type	Standard Wooden Door (g)	Mahogany Door (g)	Molded Door (g)	Average Threshold (g)
Knocking	14.2	15.8	15.0	15.0
Wiggling	0.4	0.6	0.5	0.5
Forcing	96.5	103.5	100.0	100.0

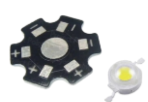
This table presents the computed threshold values for window intrusion detection calibration in the Hybrid Home Security System. The researchers conducted 10 tests in order to calibrate the system through three types of windows, namely, sliding windows, steel casement windows, and awning windows. The team conducted various interactions including knocking on, trying to wiggle, and forcing open the window in order to obtain sensor data from the MPU6050 sensor. These were taken as average values, and these were used as threshold values within the system program.

APPENDIX D. Project Costing

Hybrid Home Security System

	Materials	Item Price	Quantity	Over All Price
	Raspberry Pi 5	PHP 5,618	1 piece	PHP 5,618
	Raspberry Pi Zero 2W	PHP 2,066	1 piece	PHP 2,066
	Raspberry Pi Camera Module 3	PHP 2,631	1 piece	PHP 2,631
	ESP32 Super Mini	PHP 185	1 piece	PHP 185
	Raspberry Pi 7-inch Touch Screen	PHP 1,809	1 piece	PHP 1,809
	Magnetic Reed Switches	PHP 136	1 piece	PHP 136
	MPU6050	PHP 154	1 piece	PHP 54
	PIR HC-SR501	PHP 62	1 piece	PHP 62

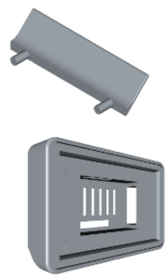
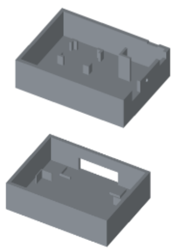
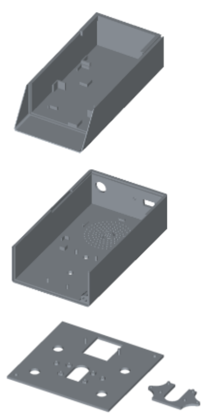
	8ohm 3w Speaker	PHP 129	1 piece	PHP 129
	Lithium-ion Battery	PHP 55	4 pieces	PHP 220
	IP2312	PHP 73	1 piece	PHP 73
	TP4056 Module	PHP 30	1 piece	PHP 30
	BMS 1S 12A	PHP 57	1 piece	PHP 57
	MT3608	PHP 24	1 piece	PHP 24
	TPS61088	PHP 96	1 piece	PHP 96
	MAX17043	PHP 225	1 piece	PHP 225

	Power Supply	PHP 1,075	1 piece	PHP 1,075
	Nut Female DC Power	PHP 8	1 piece	PHP 8
	Power Adapter	PHP 110	1 piece	PHP 110
	Red LED	PHP 25	2 pieces	PHP 50
	1w LED Bead	PHP 19	4 pieces	PHP 76
	PAM8403	PHP 75	1 piece	PHP 75
	Web Camera	PHP 950	1 piece	PHP 950
	USB Microphone	PHP 113	1 piece	PHP 113
	USB Aux mic Adapter	PHP 189	1 piece	PHP 189

	Aux Audio Cable	PHP 79	1 piece	PHP 79
	330 ohms resistor	PHP 4	2 pieces	PHP 8
	10k ohms resistor	PHP 4	2 pieces	PHP 8
	1 ohm 0.5w resistor	PHP 5	4 pieces	PHP 20
	Low ESR Capacitor	PHP 38	1 piece	PHP 38
	1000 uF 16v Capacitor	PHP 37	1 piece	PHP 37
	SS54 5A Diode	PHP 12	2 pieces	PHP 24
	LD06AJSA	PHP 61	1 piece	PHP 61
	Stranded Wires	PHP25	10 cm	PHP 250
	MicroSD Card	PHP 365	1 piece	PHP 365

	Universal PCB	PHP 60	1 piece	PHP 60
	Pin Header	PHP 20	2 pices	PHP 40
	Active Cooler	PHP 180	1 piece	PHP 180
	OTG Micro USB	PHP 67	1 piece	PHP 67
	Rocker Switch	PHP 20	2 pieces	PHP 40
	Battery Holder	PHP 75	2 pieces	PHP 150
TOTAL				**PHP 17, 488**

Other Expenses

	Materials	**Item Price**	**Quantity**	**Overall Price**
	Raspberry Pi 5 and 7-inch Touchscreen Case	PHP 1,500	1 piece	PHP 1,500
	Contact Detection Module Case	PHP 534	1 piece	PHP 534
	Camera Module Case	PHP 1,864.72	1 piece	PHP 1,864.72
	Wall Mount Camera Support	PHP 129	1 piece	PHP 129
	Thread Knurled Brass	PHP 13	4 pieces	PHP 52
	Weather Resistant Command Strip	PHP 55	2 pieces	PHP 110
TOTAL				**PHP 4,189.72**

Grand Total

Materials	PHP 17,488
Other Expenses	PHP 4,189.72
GRAND TOTAL	**PHP 21, 677.72**

APPENDIX E. Certificate of English Critique

FERDINANDRELLOROSA
ferdinand_rellorosa@dlsu.edu.ph

CERTIFICATE OF PROOFREADING

This certification confirms that the manuscript entitled "*Hybrid Home Security System with Contact Detection and Computer Vision-Based Facial Recognition*," written by **BEA L. MINDANAO, MOISES B. ROBLES, and TRINA T. TAGULINAO** has been proofread and edited by the undersigned.

The editor endeavored to ensure that punctuation, spelling, grammar, phrasing, and sentence structure were corrected while ensuring that the research content and authors' intentions were not altered.

Signed this 11th of May 2026.

FERDINAND L. RELLOROSA, LPT, MATEL
Assistant Professor IV

APPENDIX F. Certificate of Manuscript Originality

Innovation and Technology Support Services Office (ITSSO)

SLSU ITSSO

CERTIFICATE OF MANUSCRIPT ORIGINALITY

This is to certify that the research paper entitled, **"Hybrid Home Security System with Contact Detection and Computer Vision-Based Facial Recognition"**

Title of the Manuscript

submitted by **Bea L. Mindanao, Moises B. Robles, and Trina T. Tagulinao**

Faculty / **Student-Author/s**

of the **College of Engineering**

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is an outcome of an independent and original work. The manuscript received a text

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RENATO R. MAALIW III, DIT
Director, ITSSO

Email: itsso.slsu@gmail.com
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